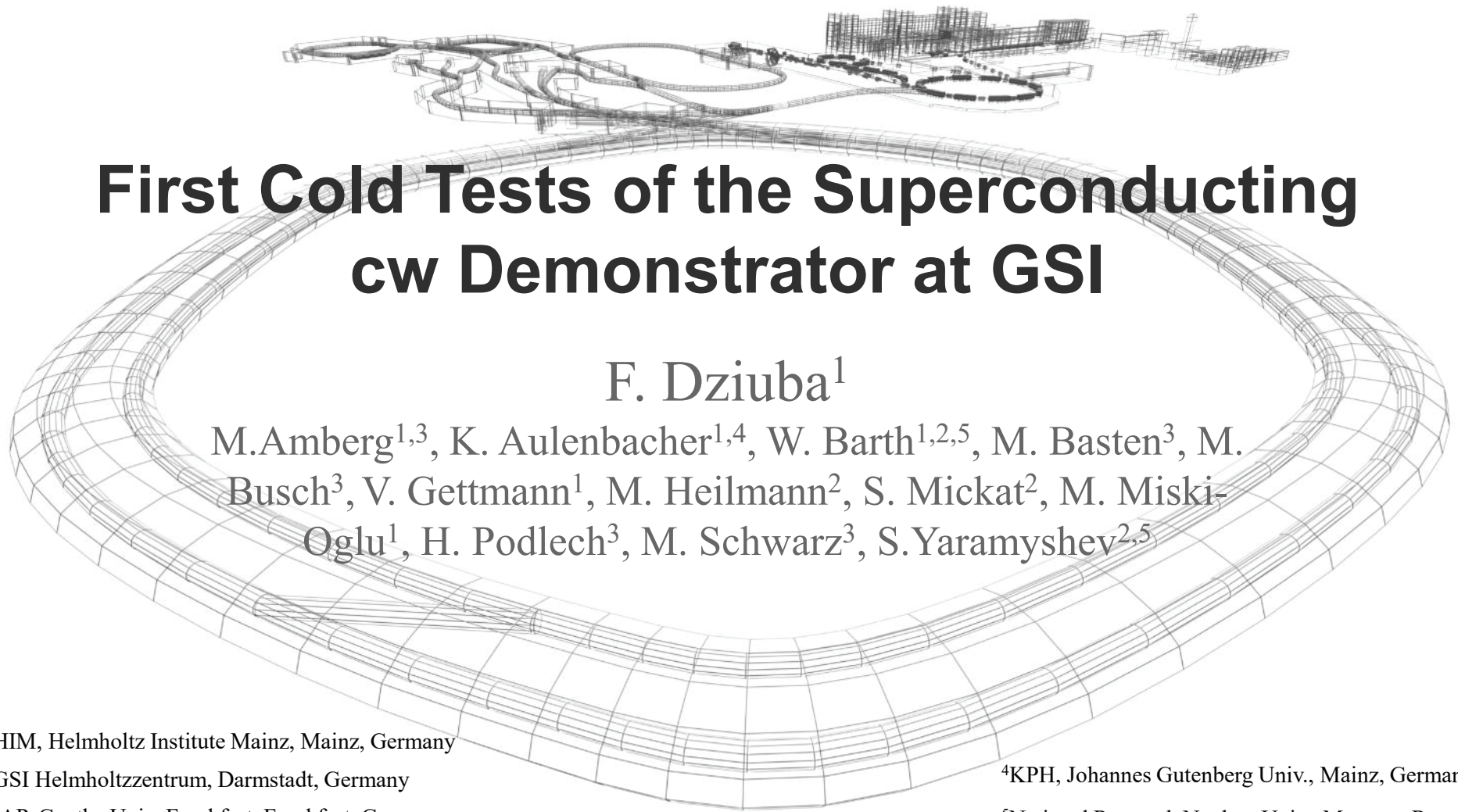




First Cold Tests of the Superconducting cw Demonstrator at GSI

F. Dziuba¹

M. Amberg^{1,3}, K. Aulenbacher^{1,4}, W. Barth^{1,2,5}, M. Basten³, M. Busch³, V. Gettmann¹, M. Heilmann², S. Mickat², M. Miski-Oglu¹, H. Podlech³, M. Schwarz³, S. Yaramyshev^{2,5}

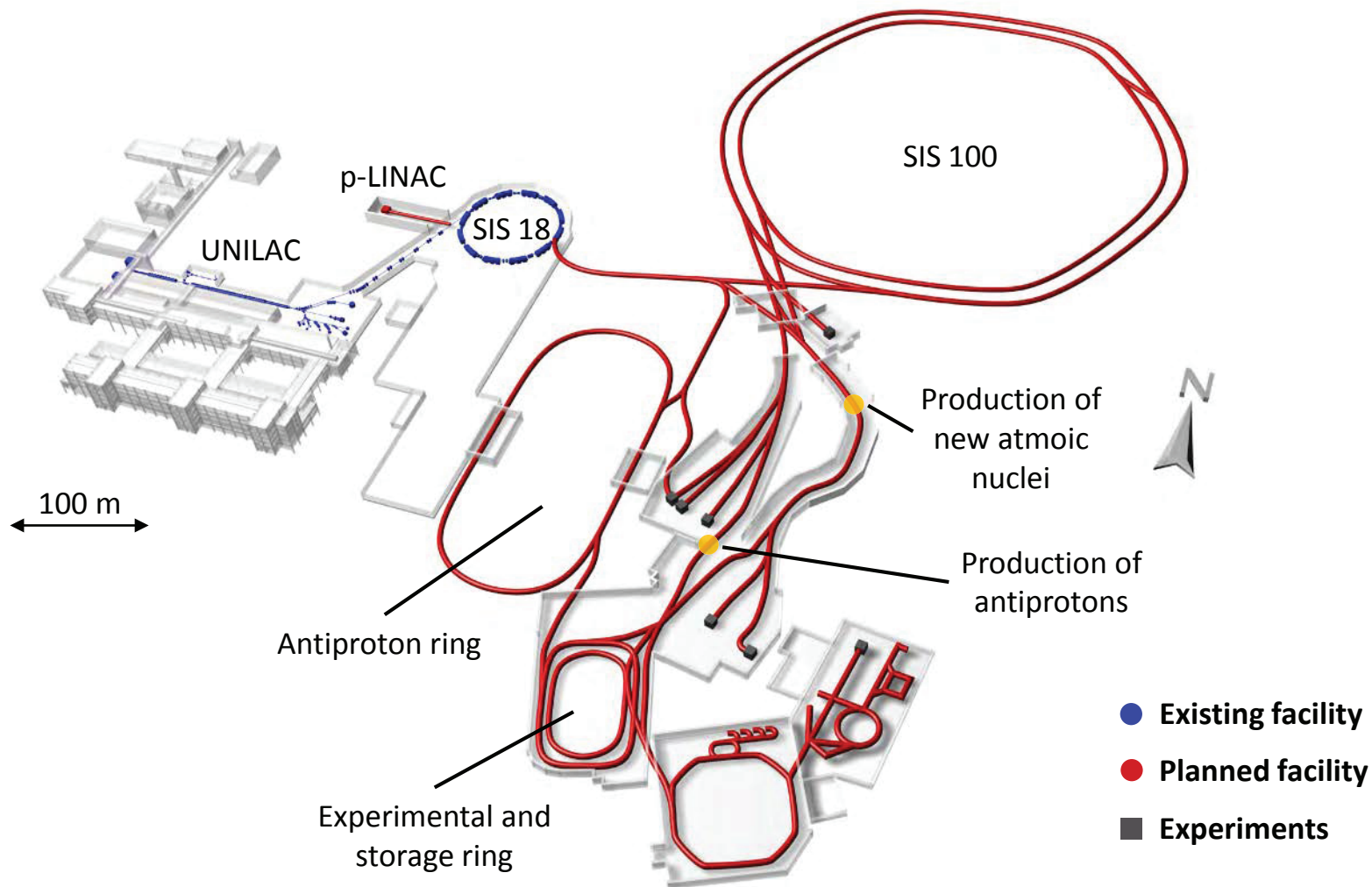
¹HIM, Helmholtz Institute Mainz, Mainz, Germany

²GSI Helmholtzzentrum, Darmstadt, Germany

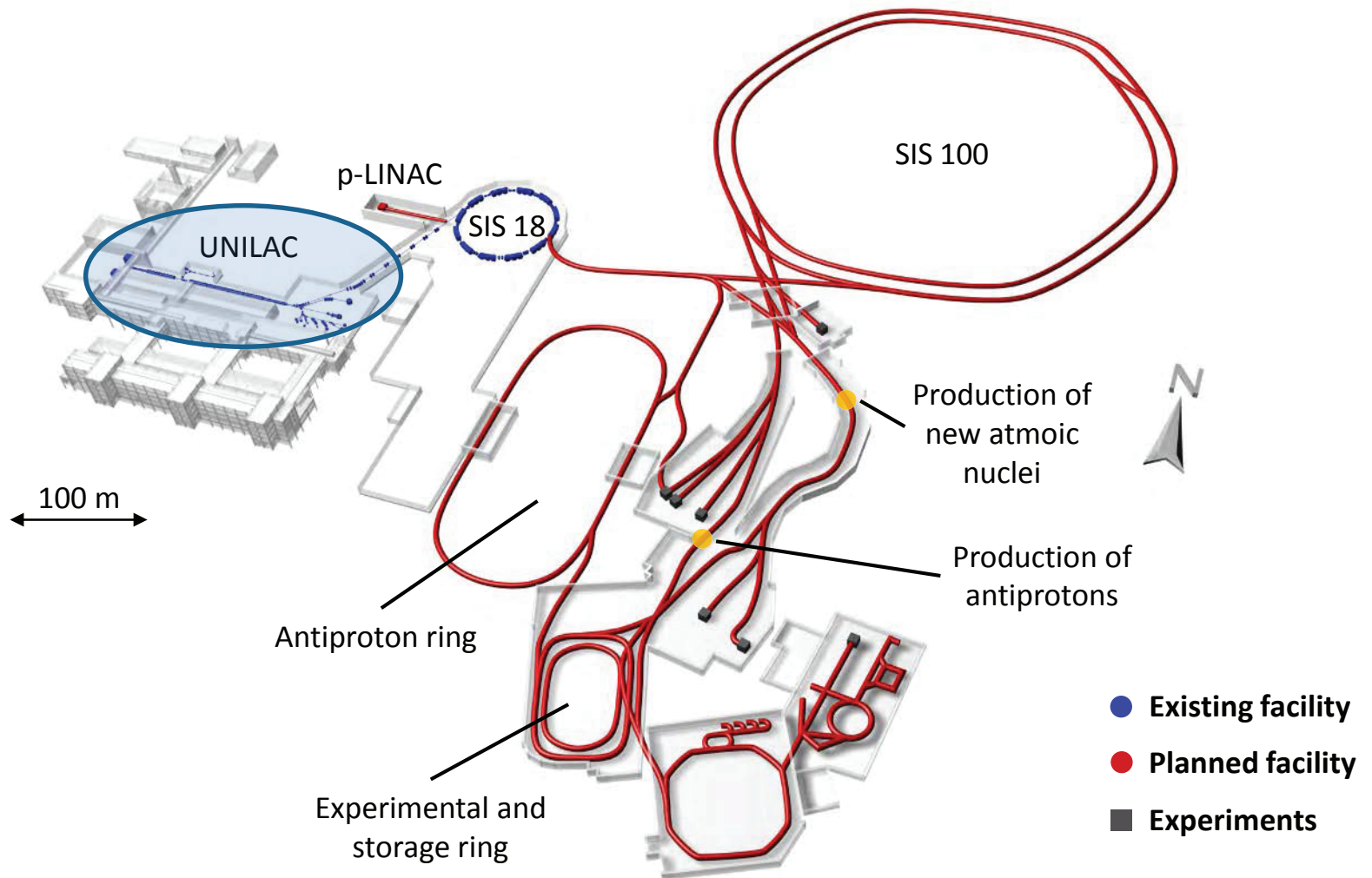
³IAP, Goethe Univ. Frankfurt, Frankfurt, Germany

⁴KPH, Johannes Gutenberg Univ., Mainz, Germany

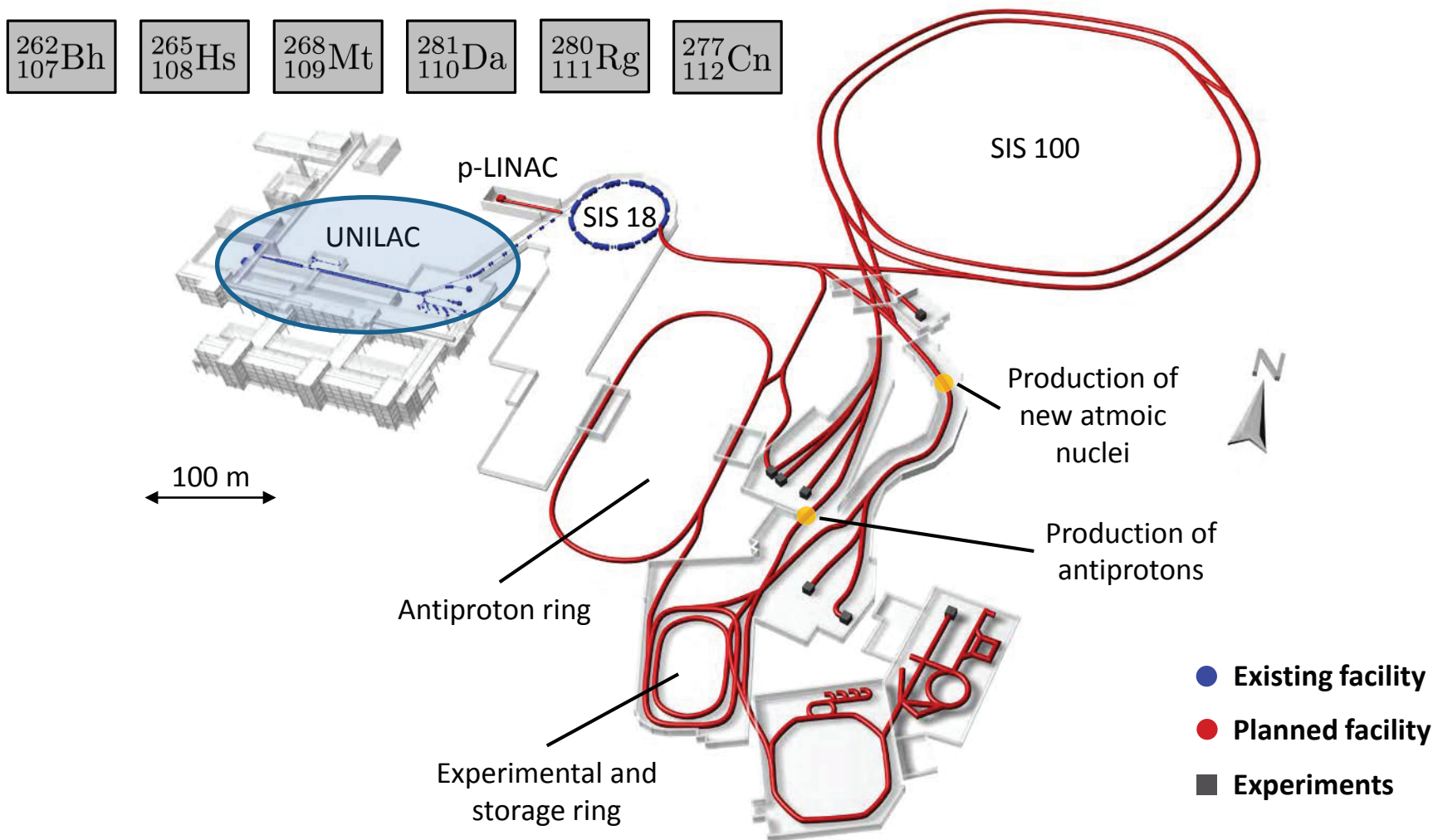
⁵National Research Nuclear Univ., Moscow, Russia



Existing GSI facility and future FAIR complex



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FAIR requirements:

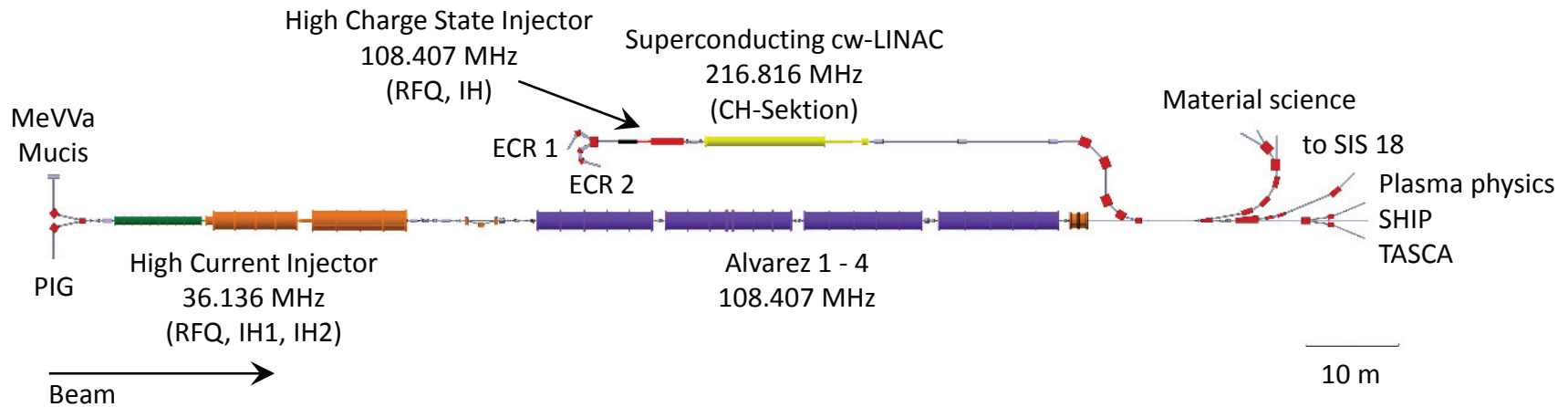
- high beam currents
- low repetition rate (max. 3 Hz)
- low duty factor (0.1 %, pulse length for SIS18 only 100 μ s)

FAIR requirements:

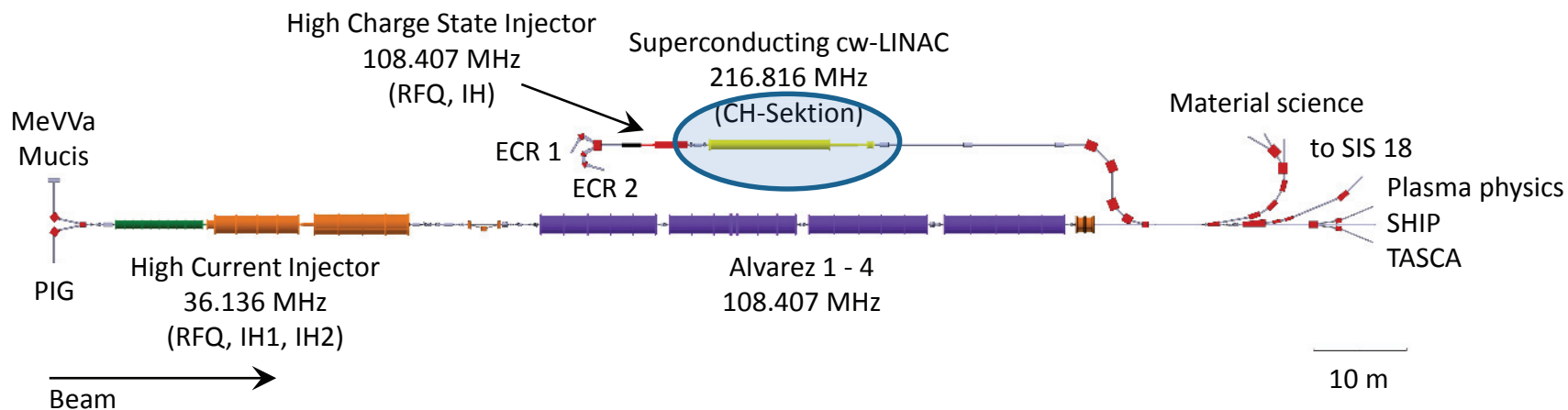
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“Super Heavy Element” requirements:

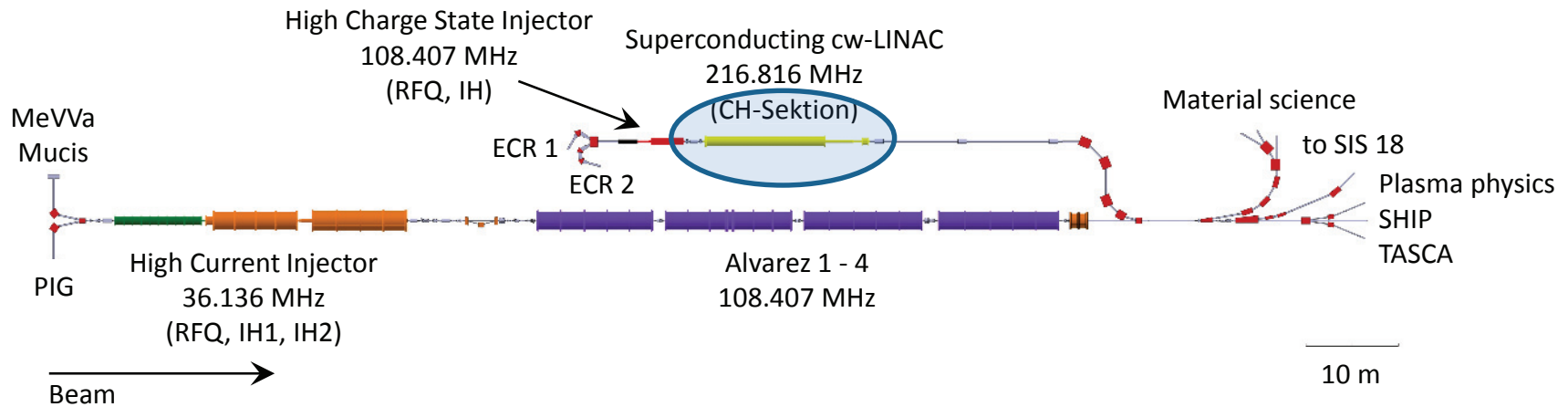
- relatively low beam currents
- high repetition rate (50 Hz)
- high duty factor (100 %, pulse length up to 20 ms)



Schematic view of the existing GSI-UNILAC with integrated superconducting cw-LINAC in parallel

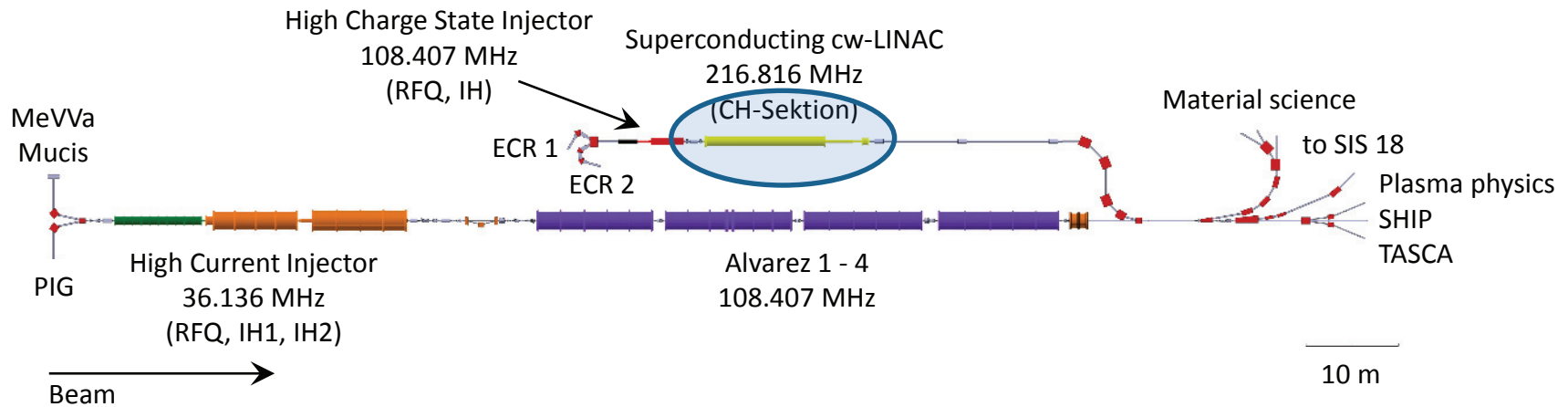


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Layout properties

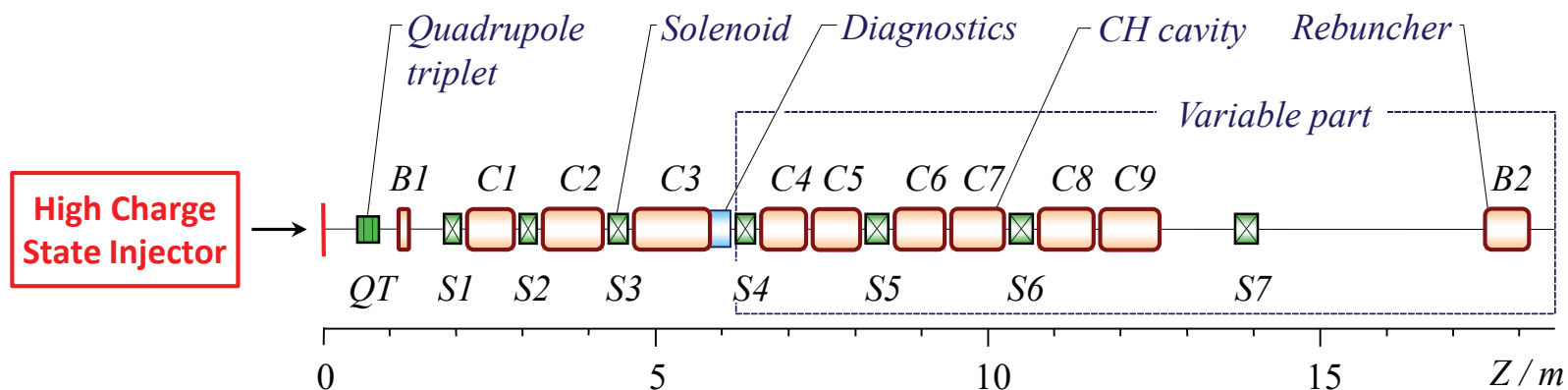


Schematic view of the existing GSI-UNILAC with integrated superconducting cw-LINAC in parallel

Layout properties

- Multigap CH cavities
- Small number of cavities with short lengths (<1 m)
- $E_a = 5.5$ MV/m enables compact linac design
- Several cavities, solenoids per cryostat
- Modular construction

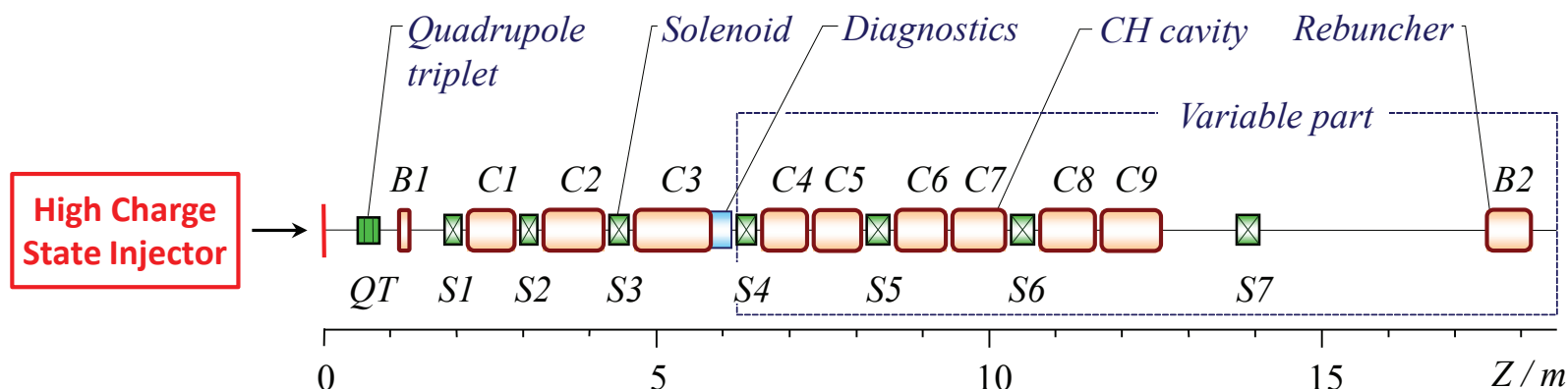
Design parameter and motivation



Parameters cw-LINAC

A/q		≤ 6
Frequency	MHz	216.816
Beam current	mA	≤ 1
Injection energy	AMeV	1.4
Output energy	AMeV	3.5–7.3
Effective length	m	13
CH cavities	#	9
Solenoids	#	7

Design parameter and motivation

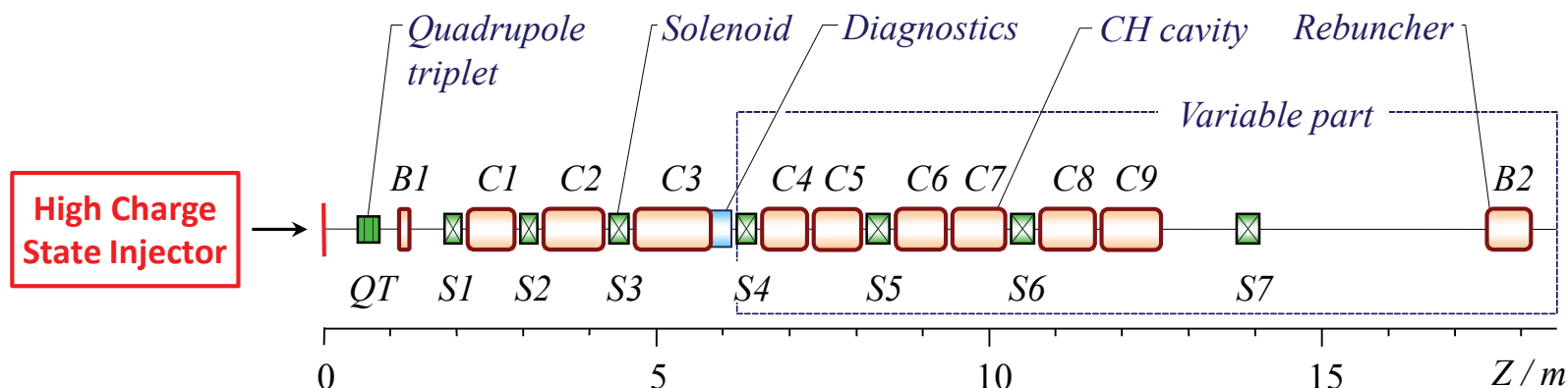


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		UNILAC	cw-LINAC
Beam intensity	particles/s	$6 \cdot 10^{12}$	$6 \cdot 10^{13}$
Beam on target	days	21	2

Design parameter and motivation



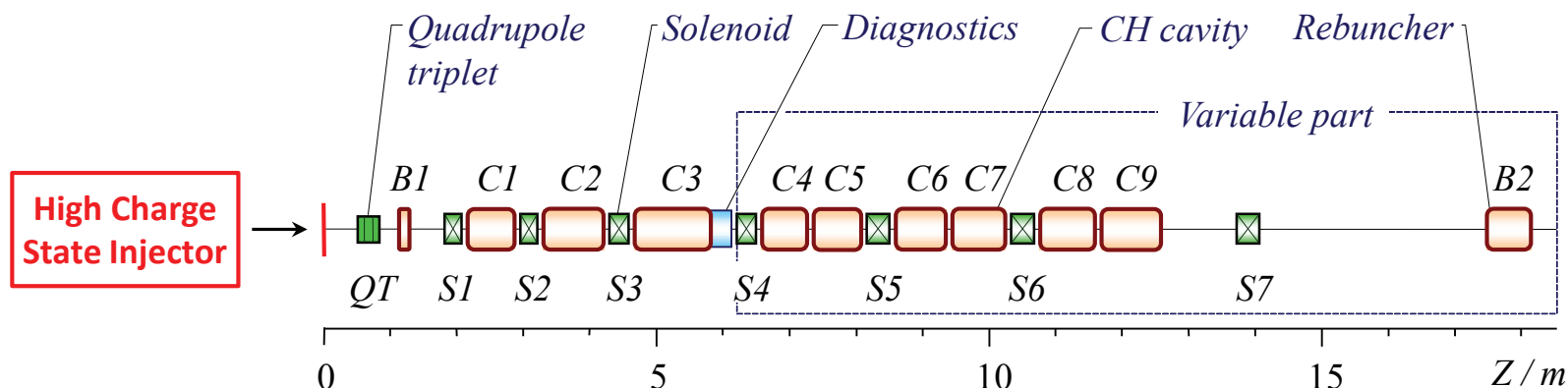
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(D. Rudolph, Lund Univ., PRL 111, 112502 (2013))

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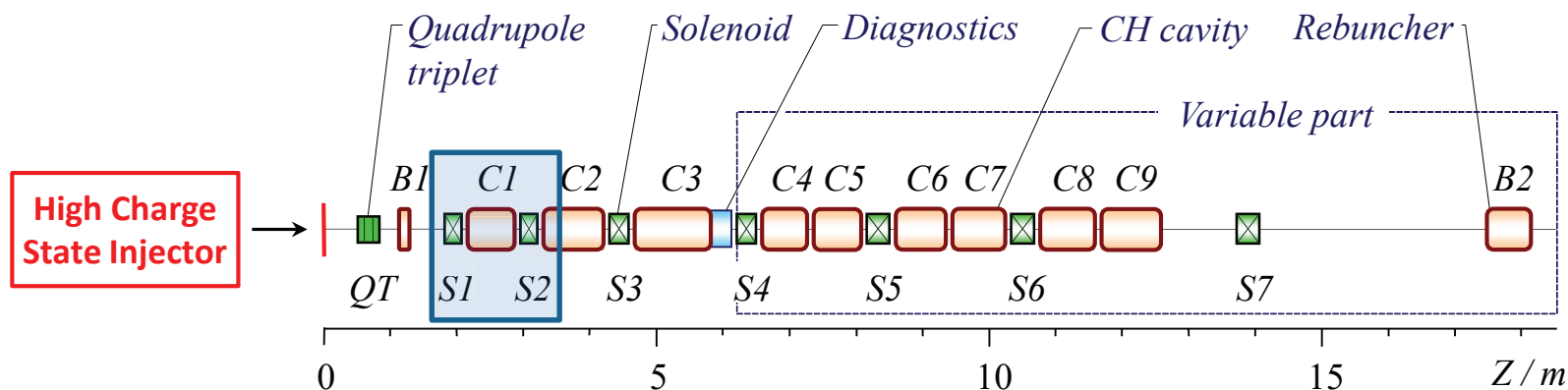
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Production of element $^{288}_{115}\text{Mc}$, $^{289}_{115}\text{Mc}$ for 30 events

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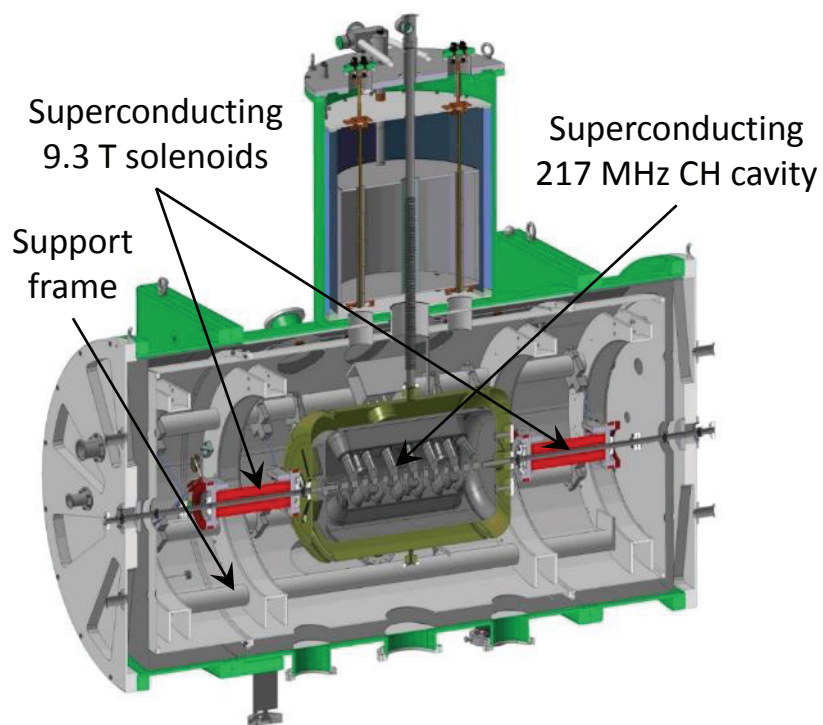
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A successful beam operation is an important milestone realizing the new superconducting cw-LINAC

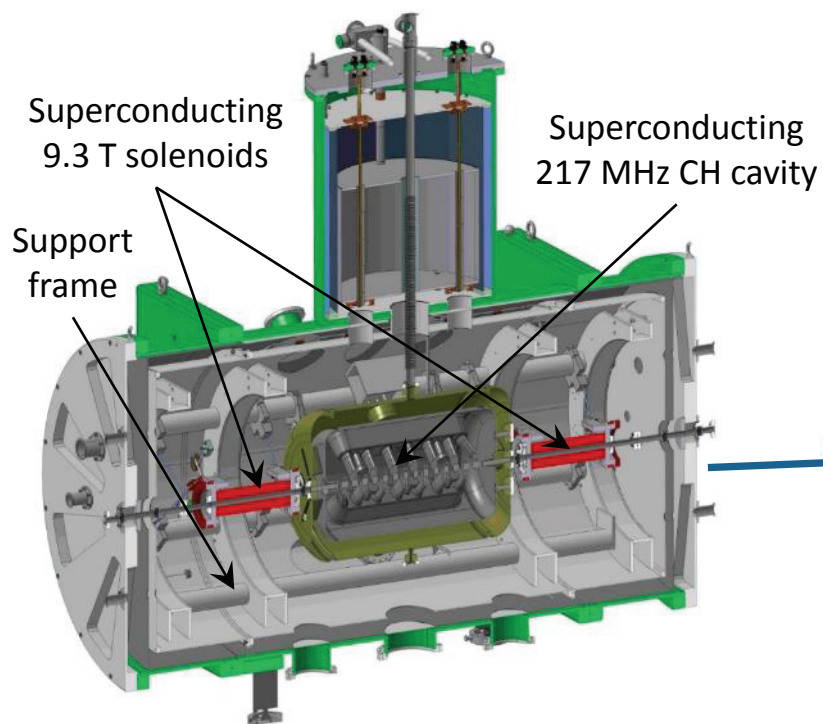


Layout of the horizontal cryomodule

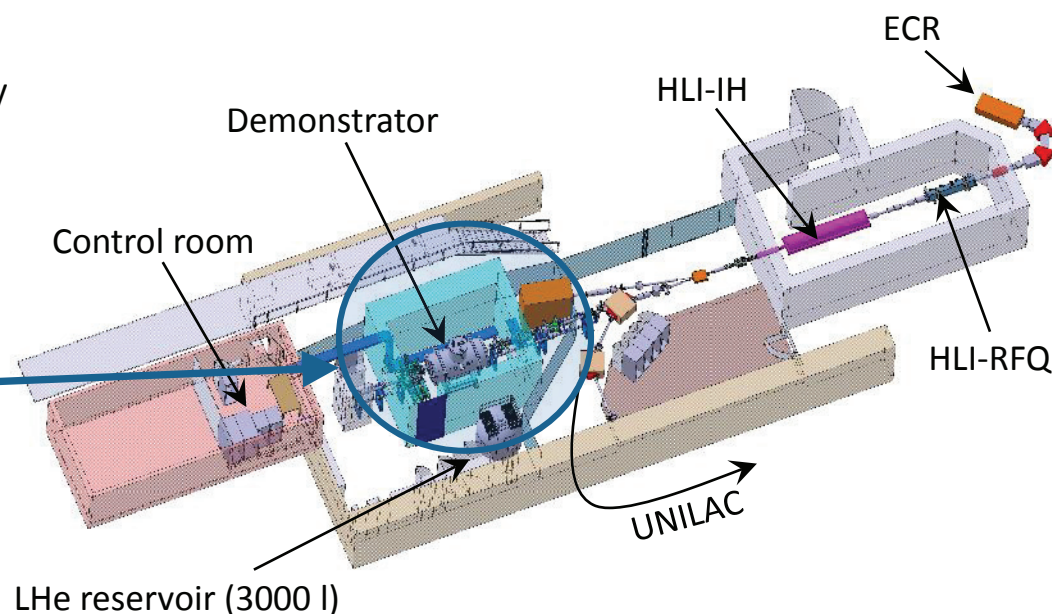
A successful beam operation is an important milestone realizing the new superconducting cw-LINAC



The High Charge State Injector (HLI) at GSI serves as an injector for the demonstrator

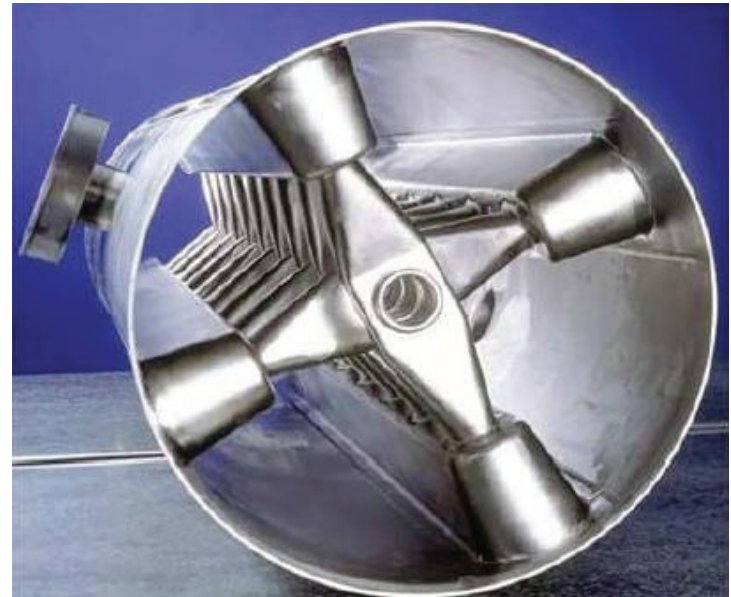
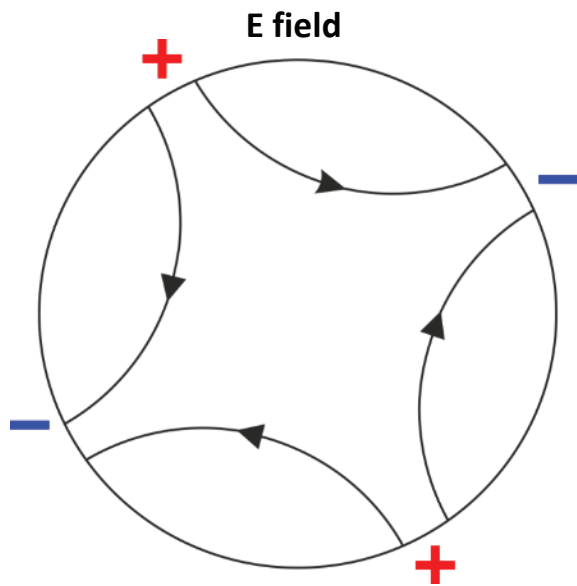


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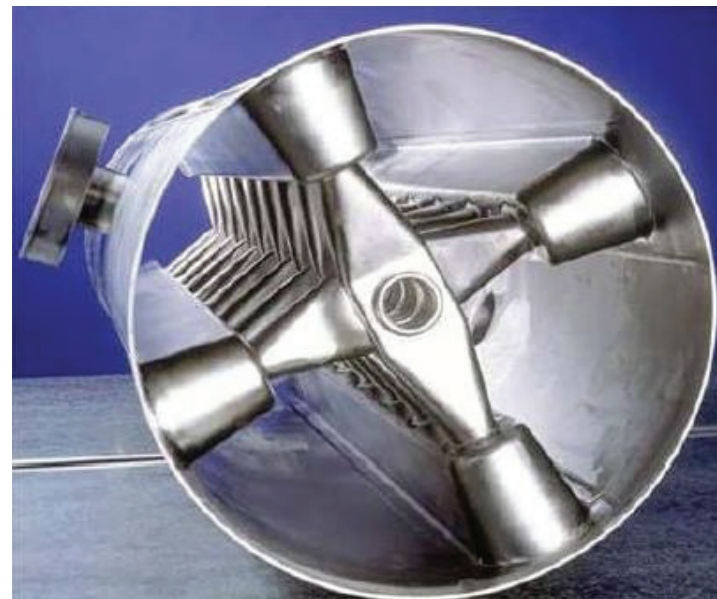
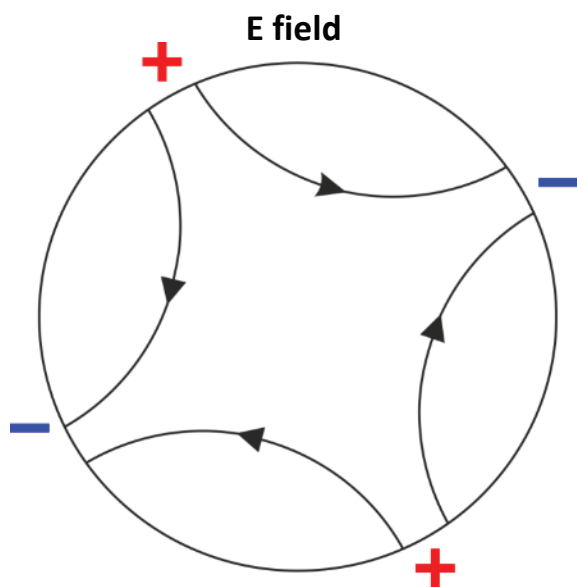


Experimental setup of the demonstrator at GSI

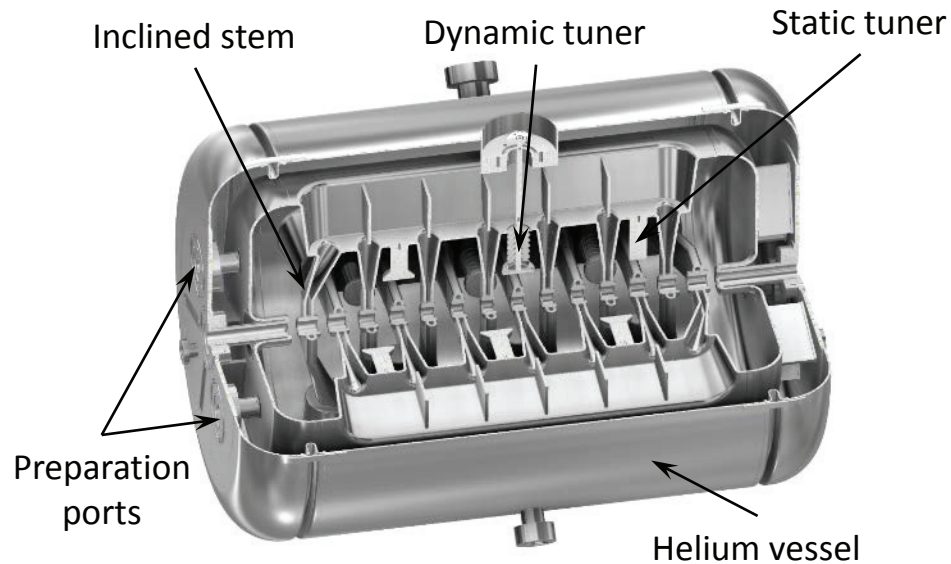
H_{211} mode of "pillbox" cavity



H_{211} mode of "pillbox" cavity

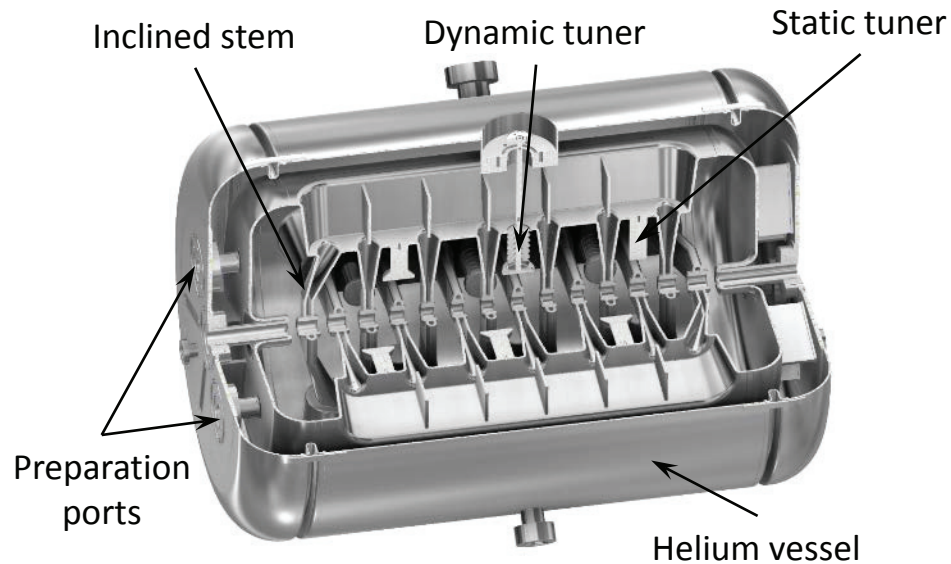


- Multigap drift tube cavity for the acceleration of protons and ions in the low and medium energy range
- Drift tubes are alternating connected to "+" and "-" potential
- **Cross-bar-H-mode** cavity → CH cavity



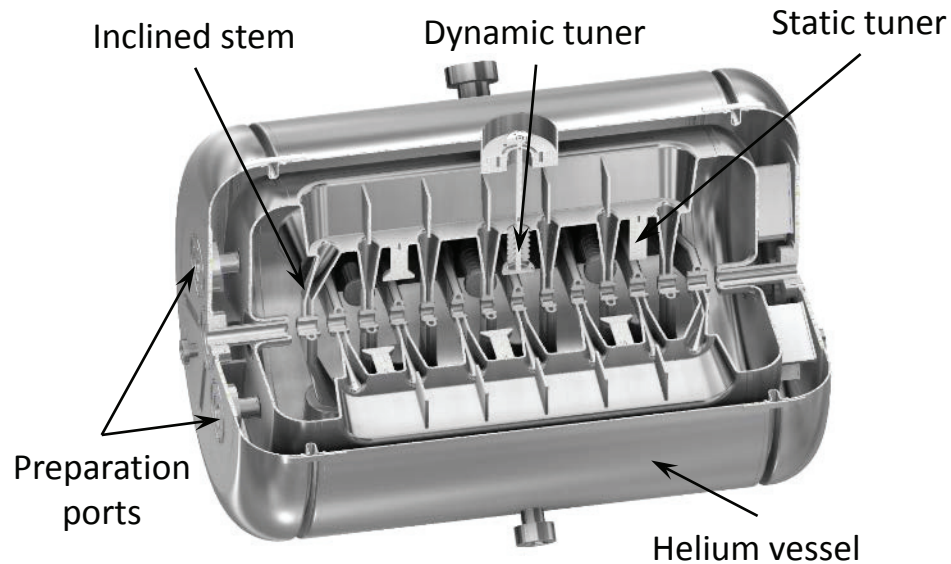
Parameters 217 MHz CH Cavity

β		0.059
Frequency	MHz	216.816
Accelerating cells		15
Effective length ($\beta\lambda$)	mm	612
Diameter	mm	409
Tube aperture	mm	18 / 20
G	Ω	52
R_a/Q_0		3240
$R_a R_S$	$k\Omega^2$	168
E_a (design)	MV/m	5.5
E_p/E_a		6.3
B_p/E_a	mT/(MV/m)	5.7



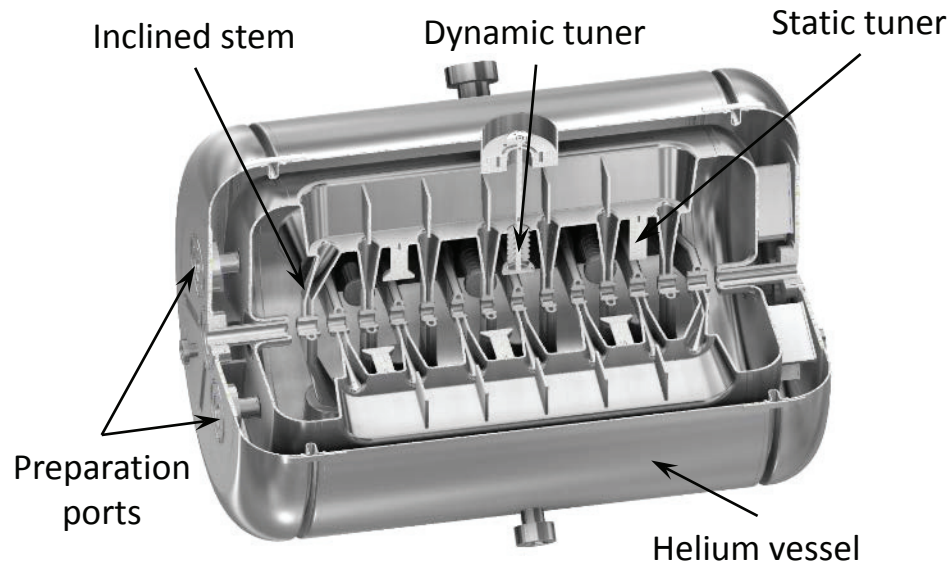
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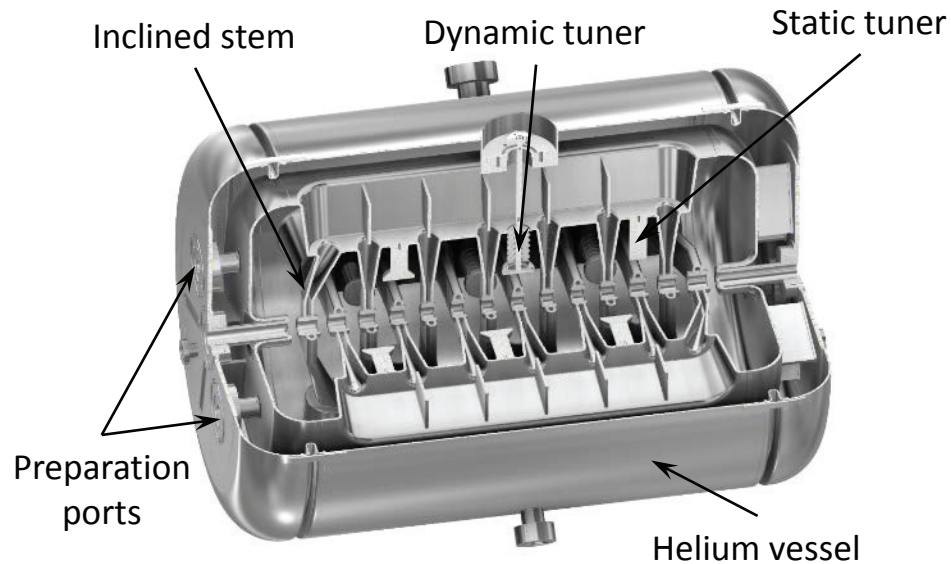
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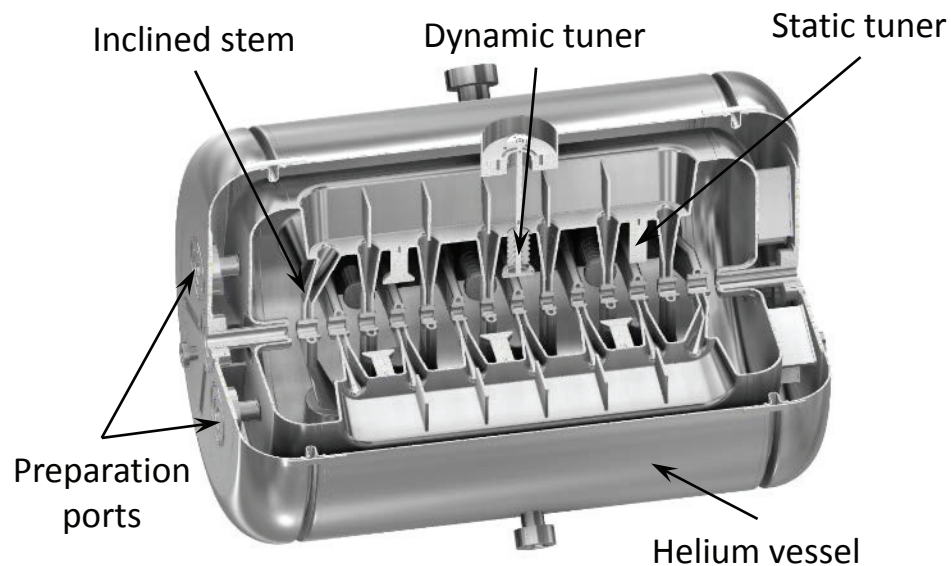
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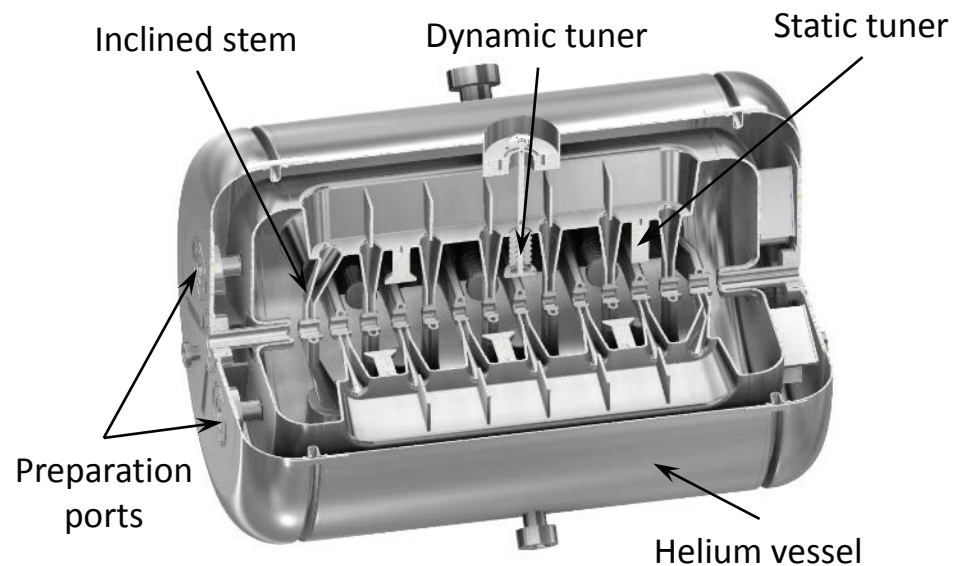
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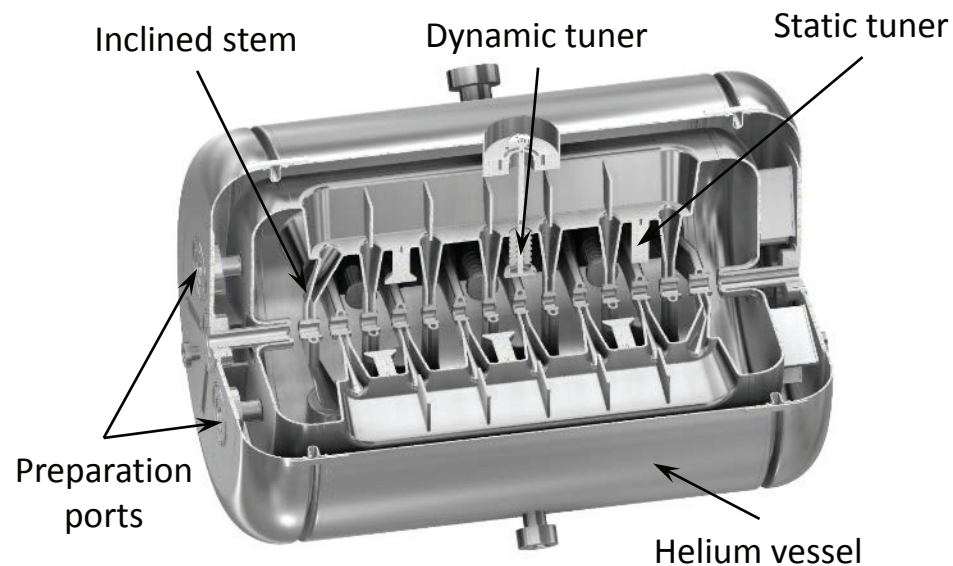
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Project start

Ordering of the cavity, solenoids and 5 kW cw RF amplifier

03/2014

Intermediate RF measurements on the cavity during fabrication

12/2015

Commissioning of cryomodule and solenoids

01/2016

First RF test of the cavity at IAP (vertical orientation)

09/2016

Assembly of He vessel to the cavity and HPR

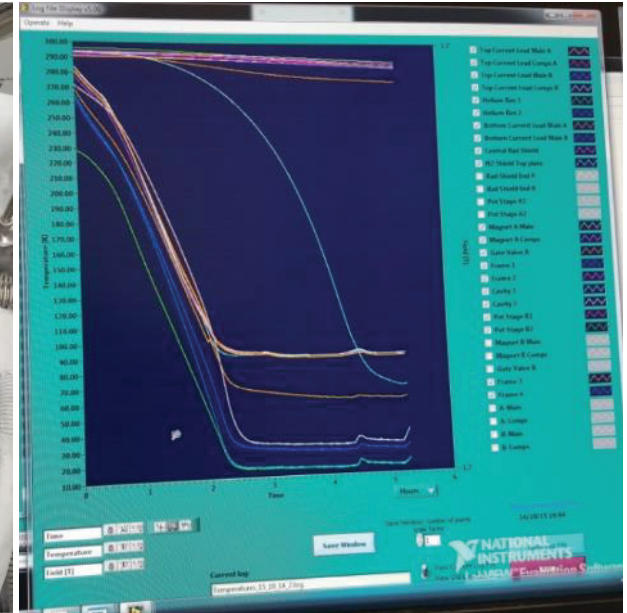
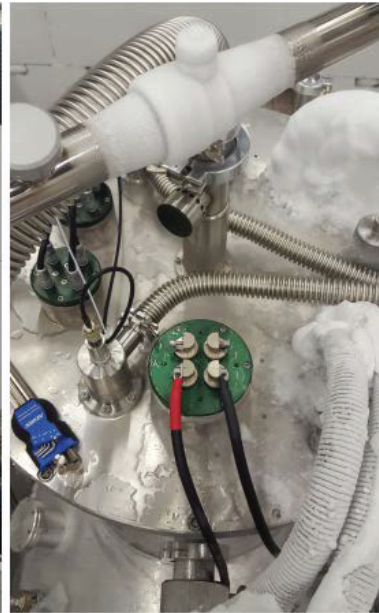
10/2016

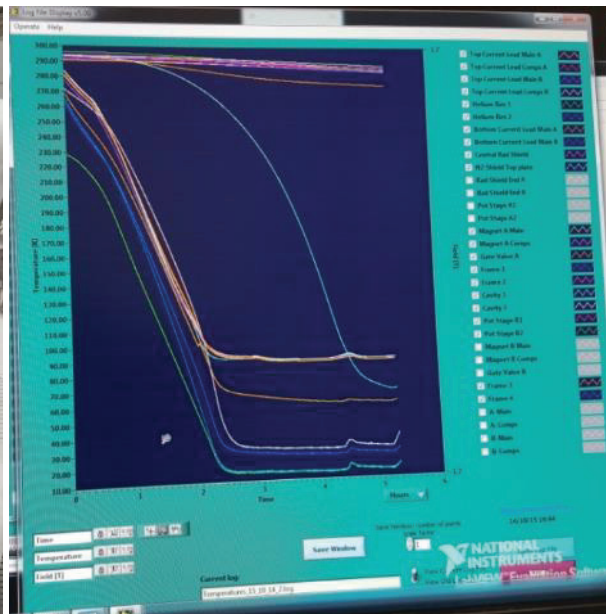
Second RF test of the cavity at GSI (horizontal orientation)

03/2017

Beam test of the demonstrator







- Superconducting solenoids were ramped at 9.3 T (0.2 A/s)
- Standby losses of cryostat 3 W
- Drift of components during cooldown ± 0.15 mm (transversal)

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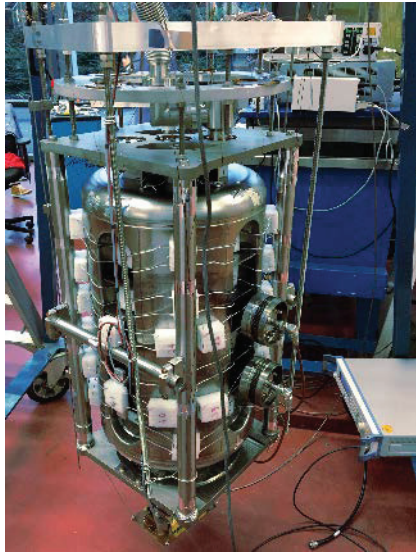
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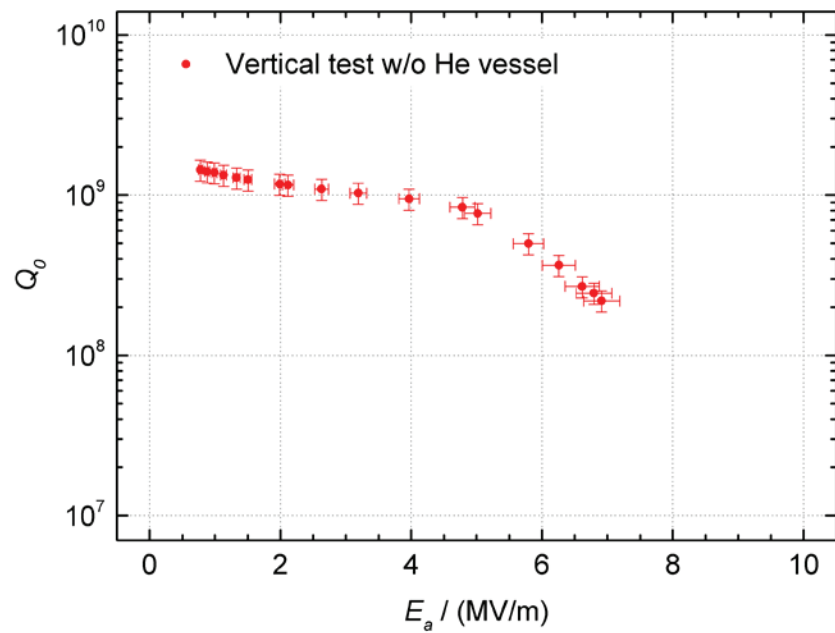
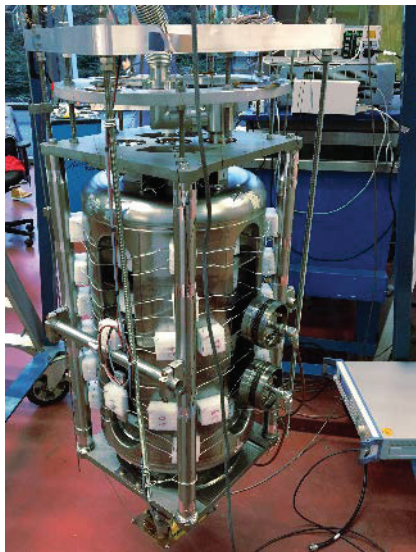
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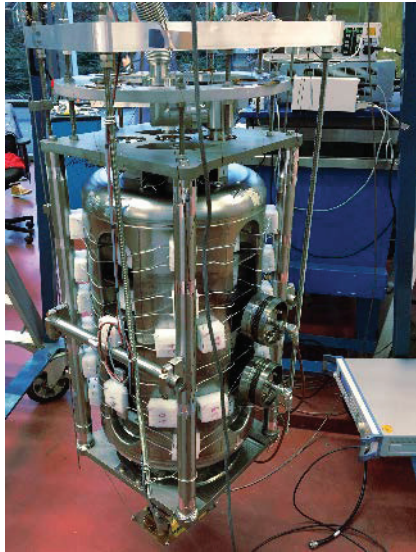
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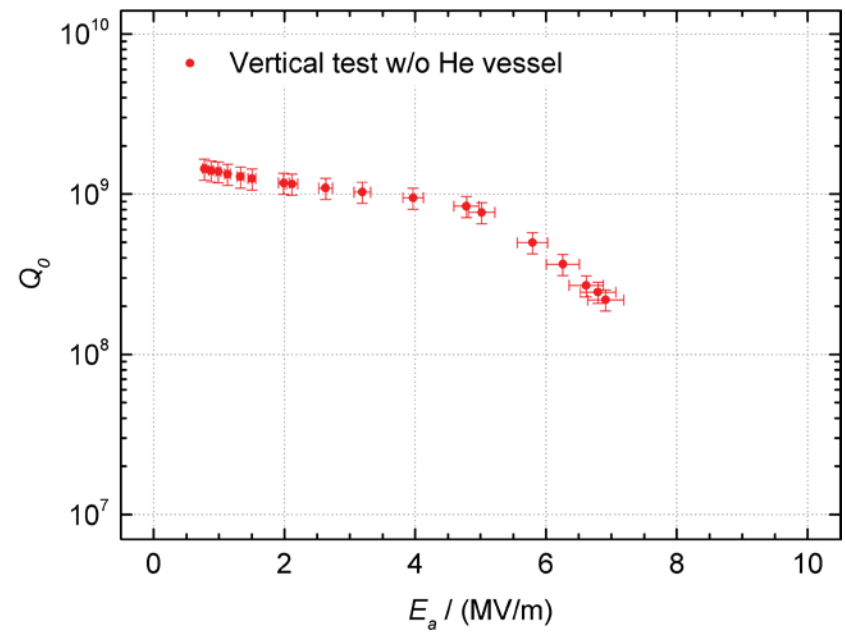


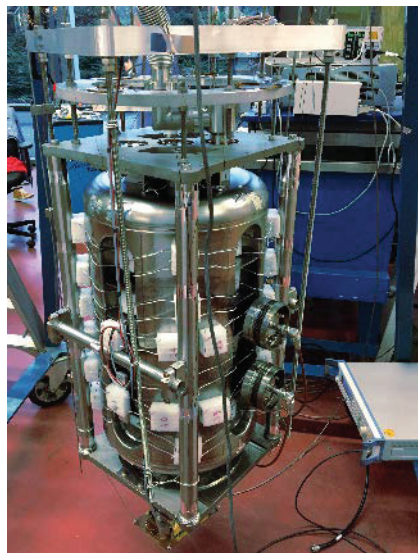




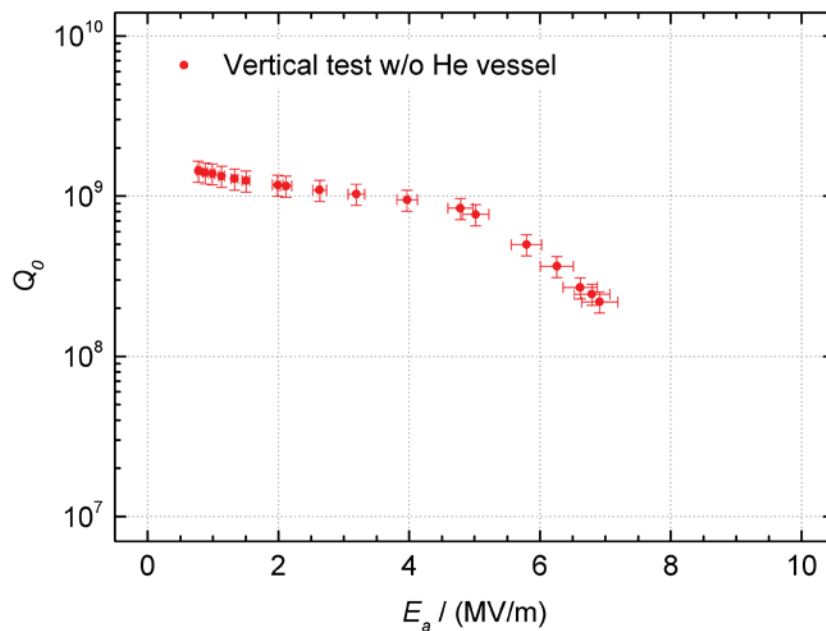


RF test in vertical orientation w/o He vessel at 4.2 K



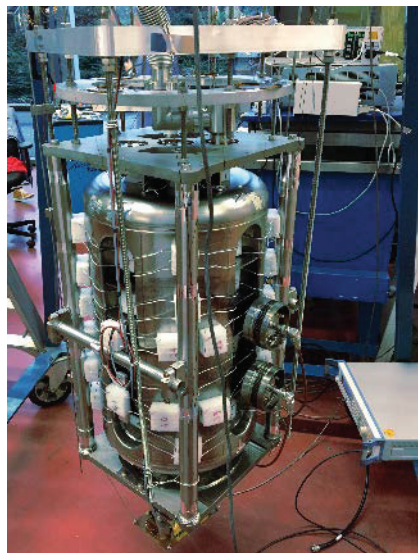


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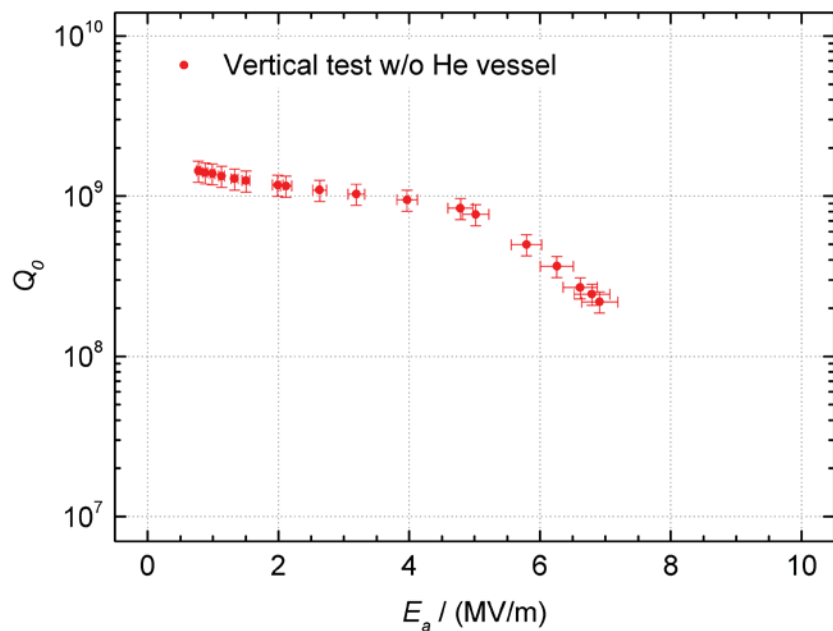


Cavity was limited by field emission due to insufficient surface preparation

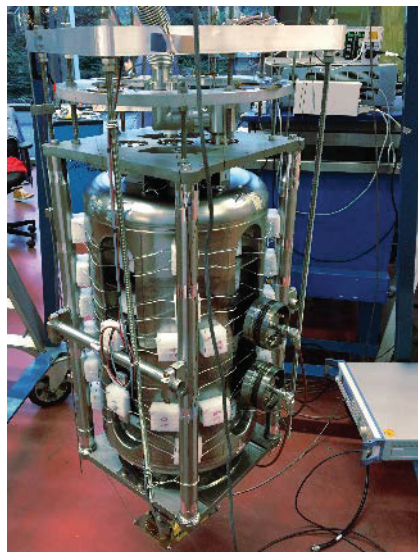
HPR was not possible inside each quadrant



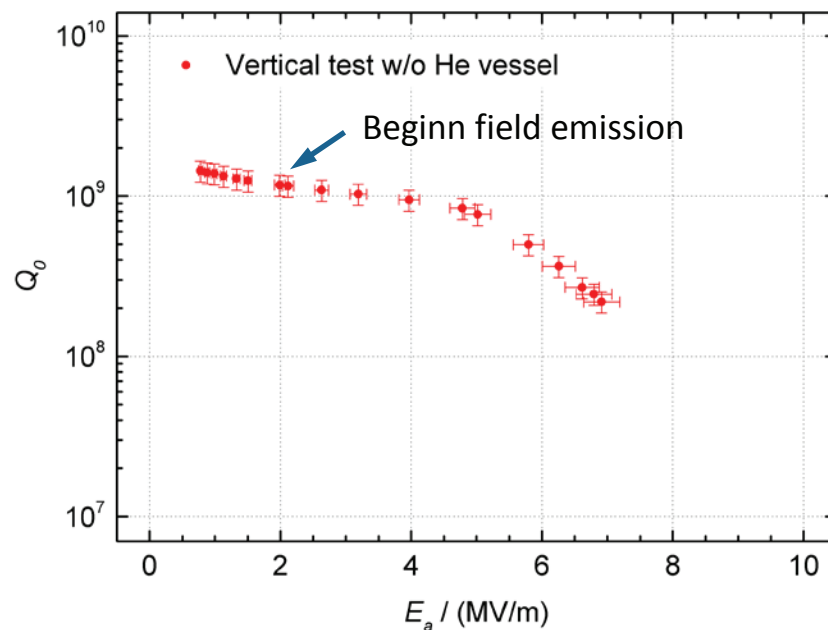
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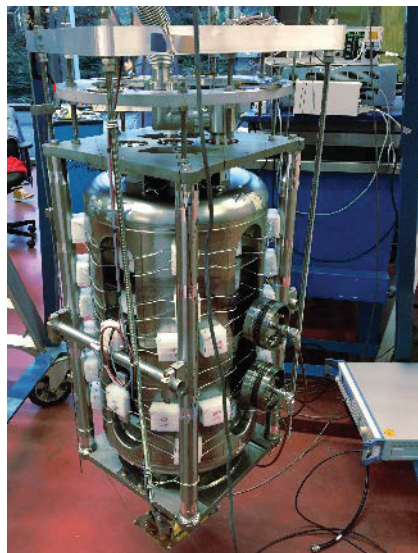
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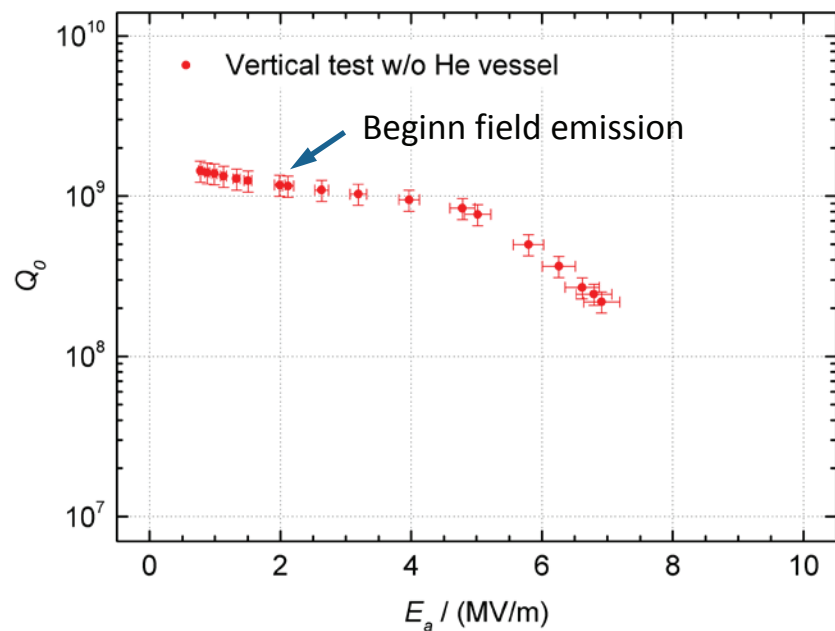
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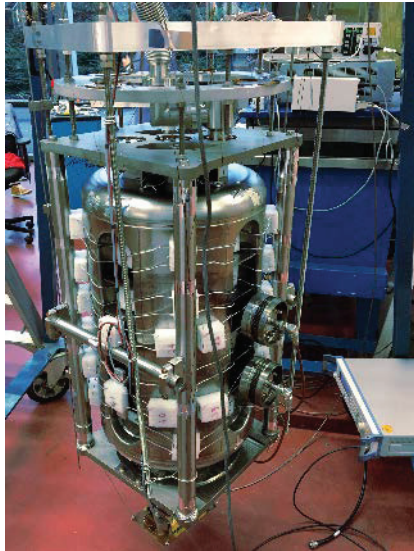
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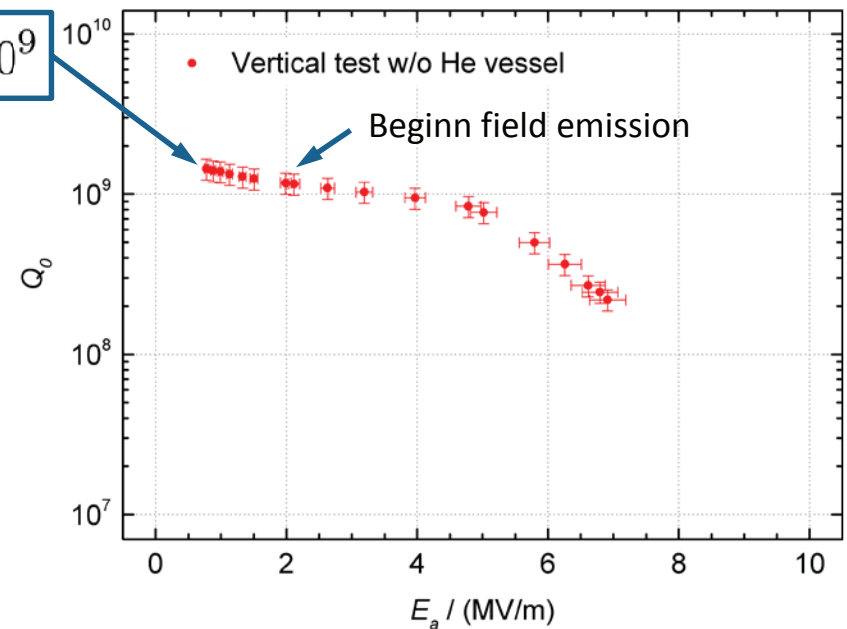


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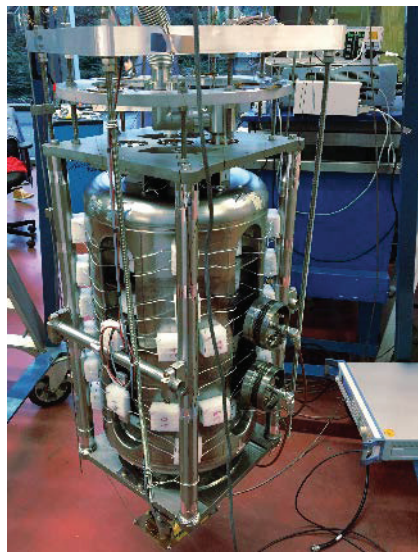


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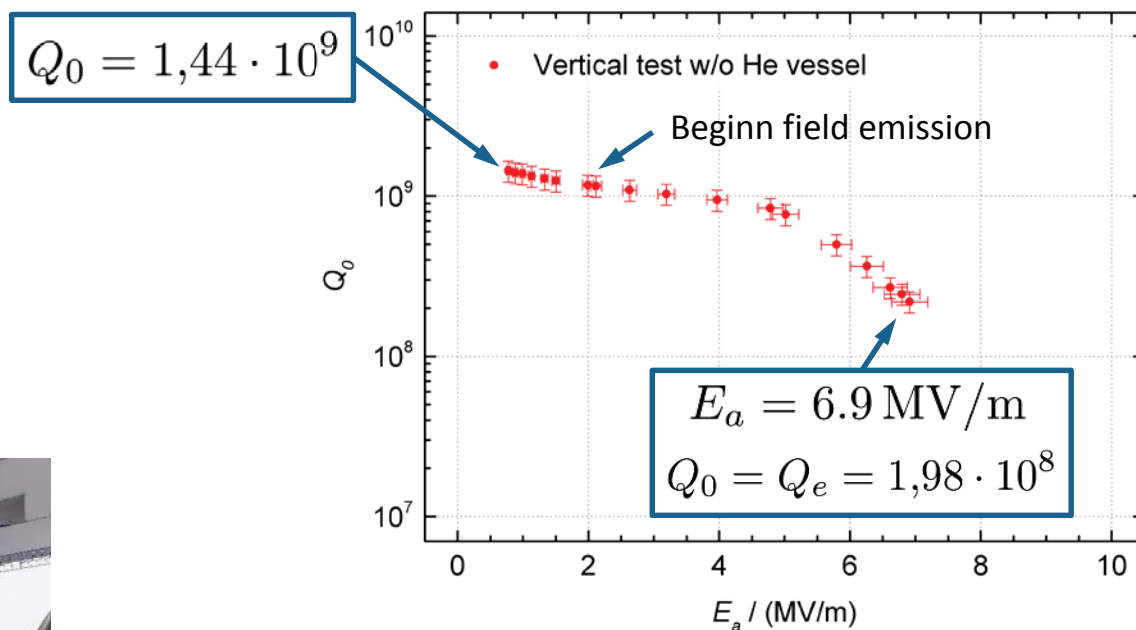
$$Q_0 = 1,44 \cdot 10^9$$



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RF test in vertical orientation w/o He vessel at 4.2 K



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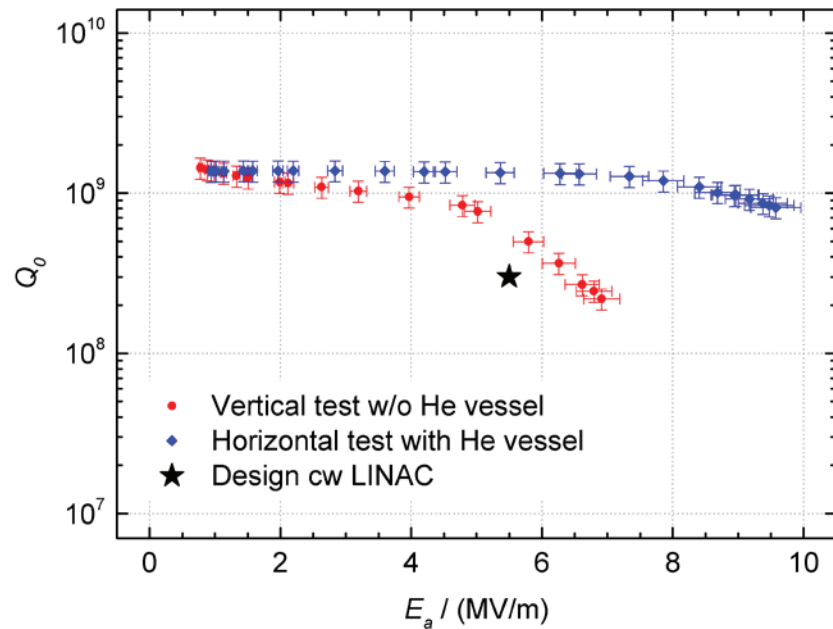
10/2016

Second RF test of the cavity at GSI (horizontal orientation)

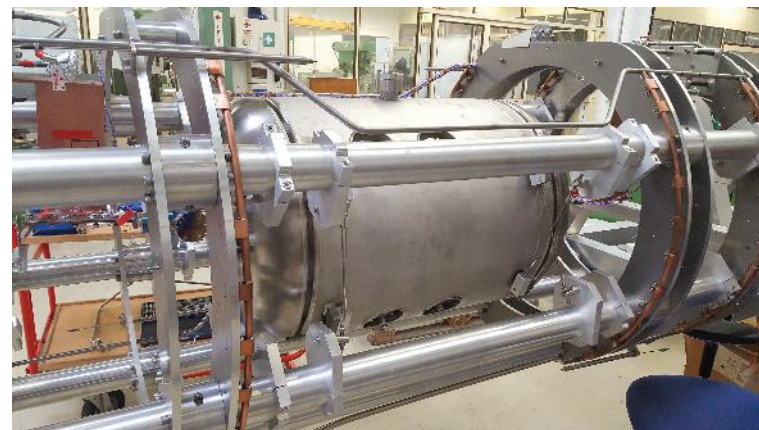
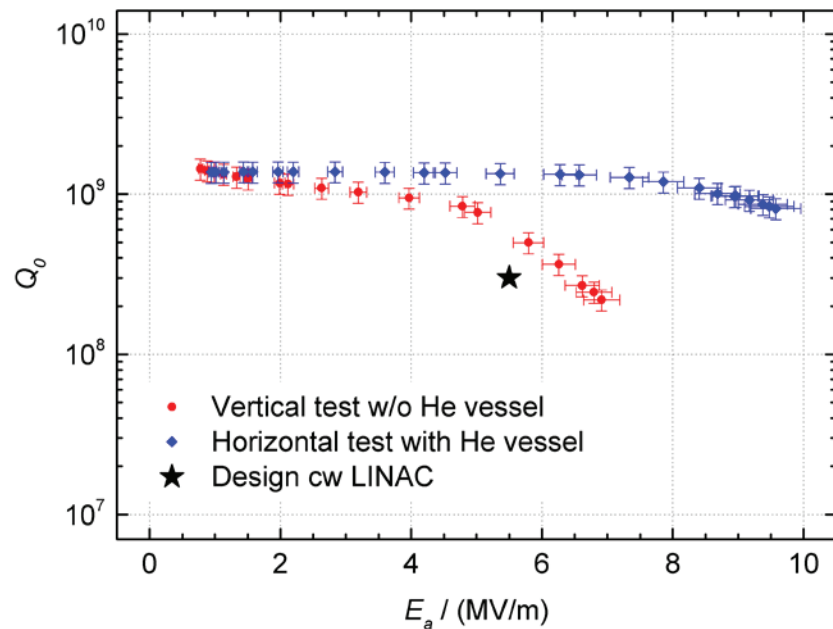
03/2017

Beam test of the demonstrator

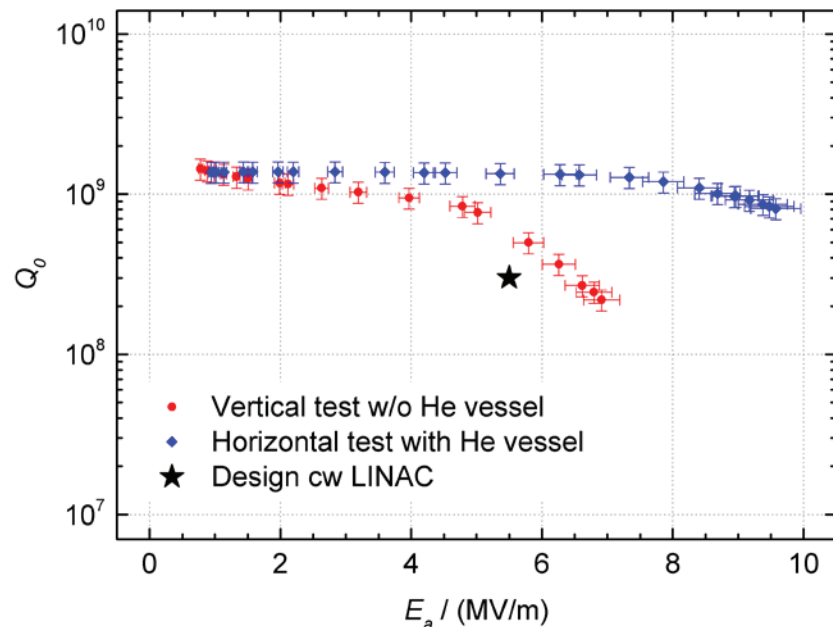




RF test in horizontal orientation with He vessel at 4.2 K



RF test in horizontal orientation with He vessel at 4.2 K



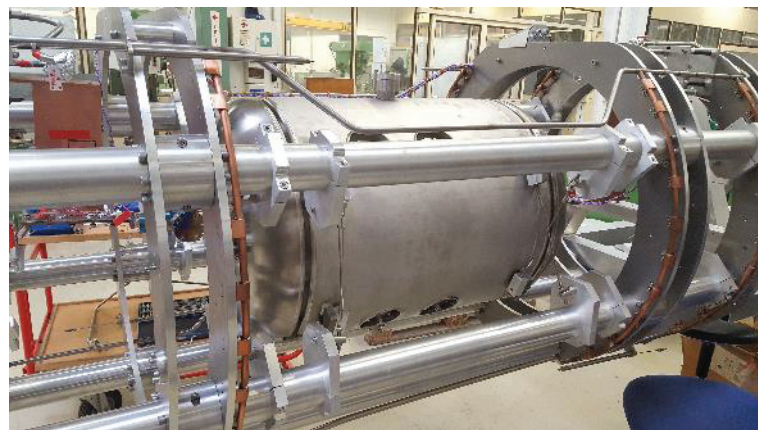
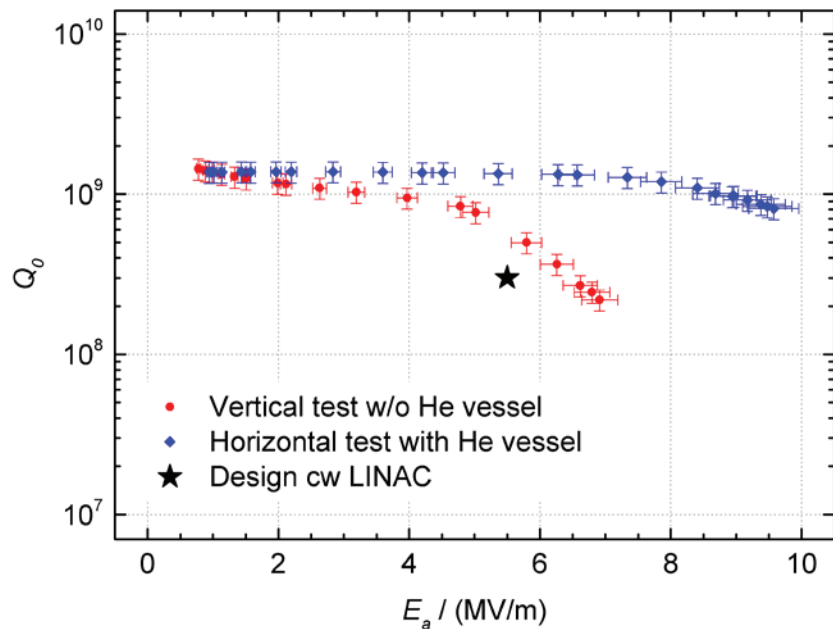
Improved performance due to an advanced HPR

Low field emission activity

Very high gradient

Thermal quenching above 9.6 MV/m

RF test in horizontal orientation with He vessel at 4.2 K



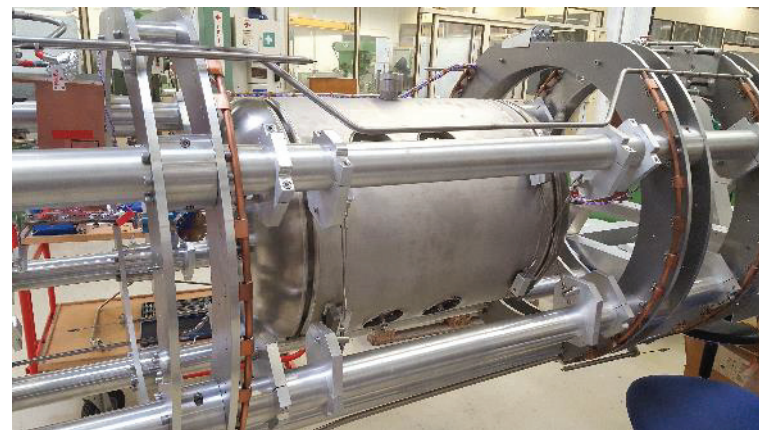
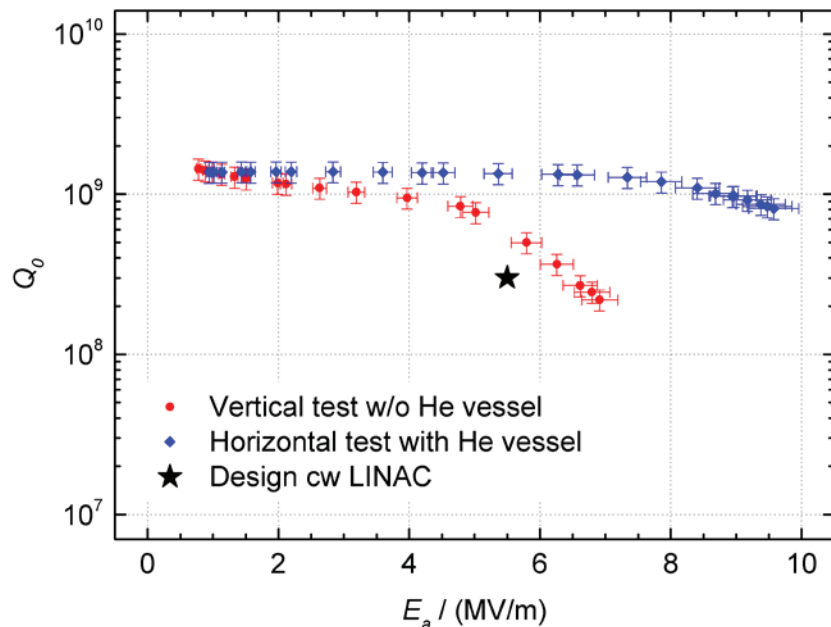
➔ Improved performance due to an advanced HPR

Low field emission activity

Very high gradient

Thermal quenching above 9.6 MV/m

RF test in horizontal orientation with He vessel at 4.2 K



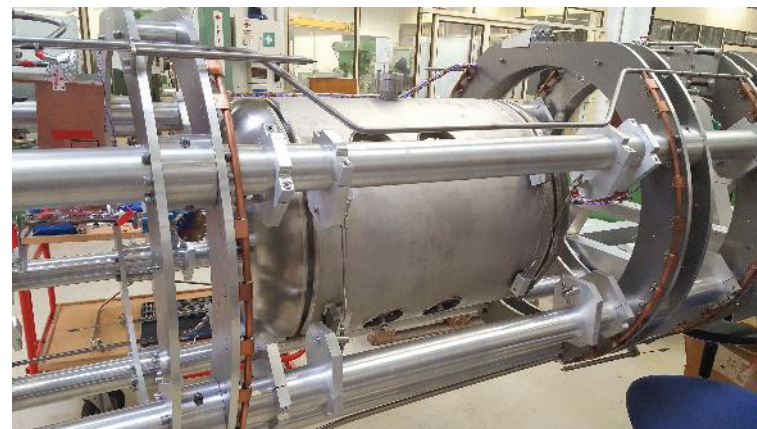
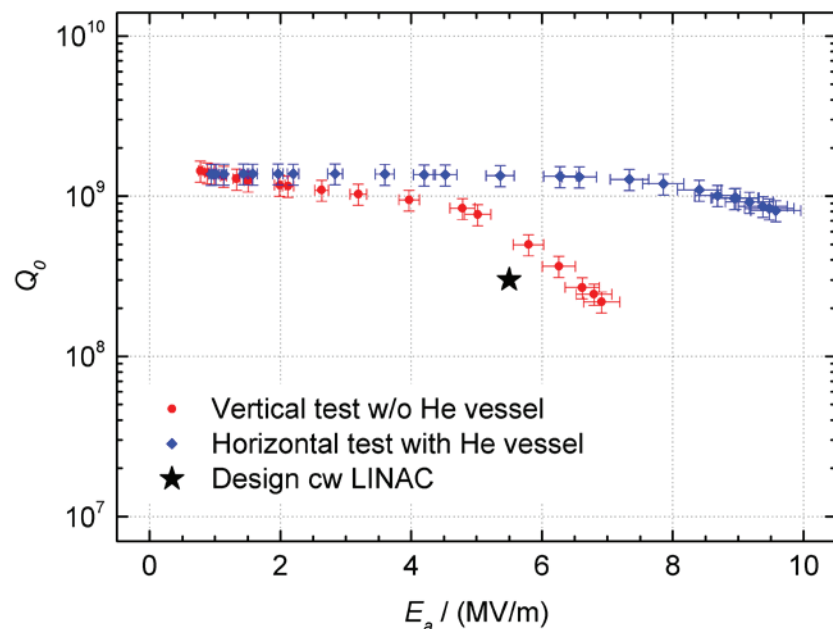
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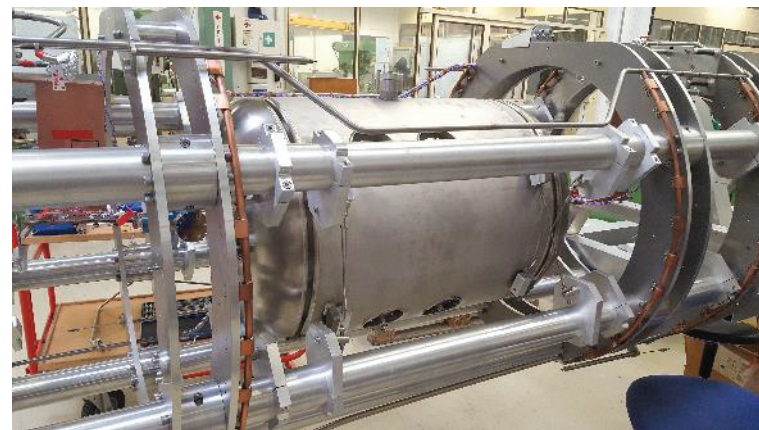
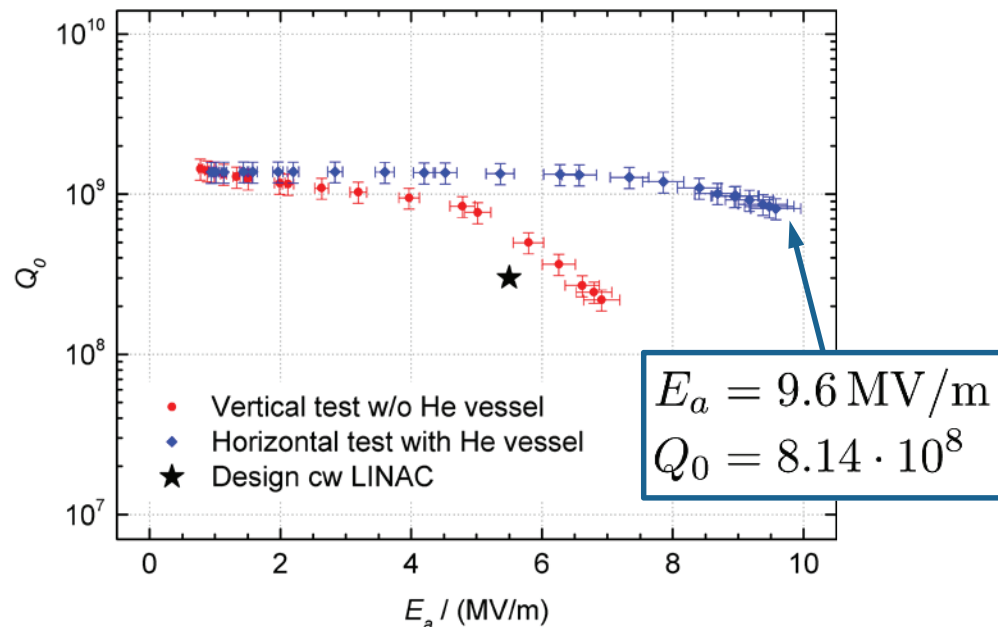
Thermal quenching above 9.6 MV/m

RF test in horizontal orientation with He vessel at 4.2 K



- ➡ Improved performance due to an advanced HPR
 - ➡ Low field emission activity
 - ➡ Very high gradient
- Thermal quenching above 9.6 MV/m

RF test in horizontal orientation with He vessel at 4.2 K



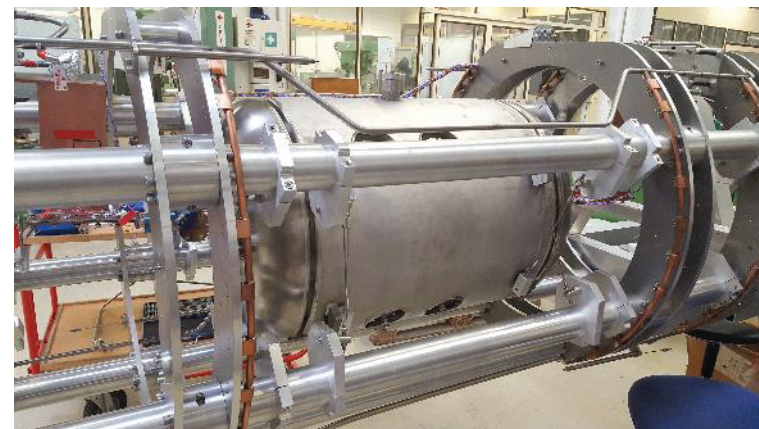
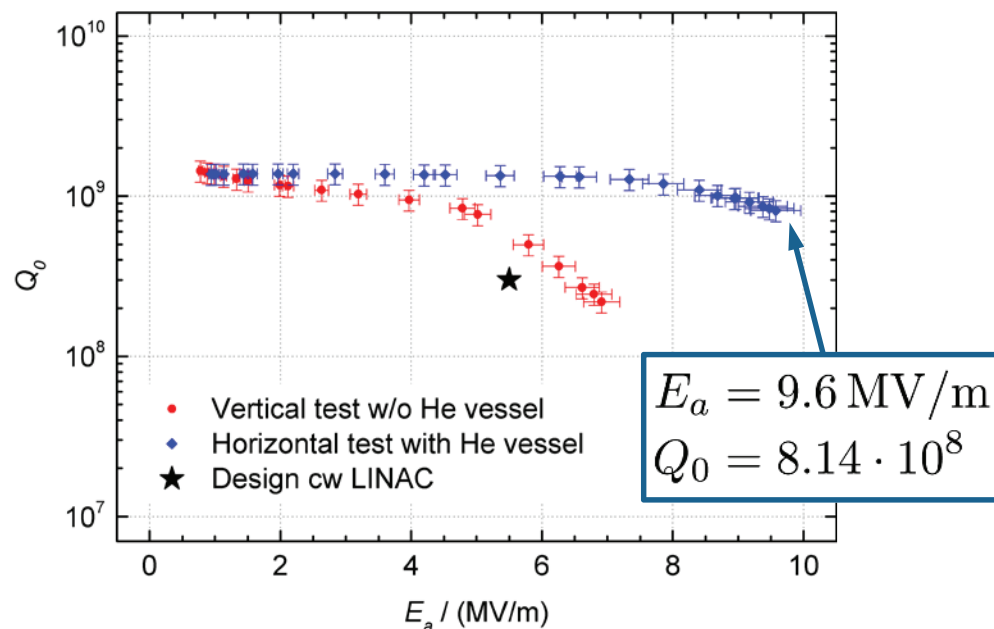
➡ Improved performance due to an advanced HPR

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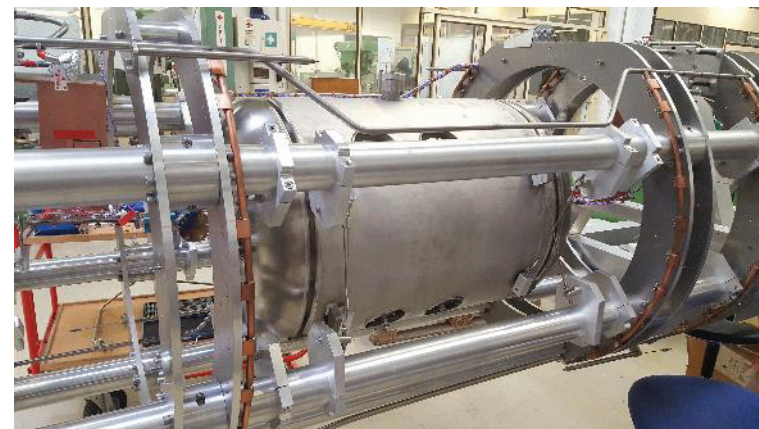
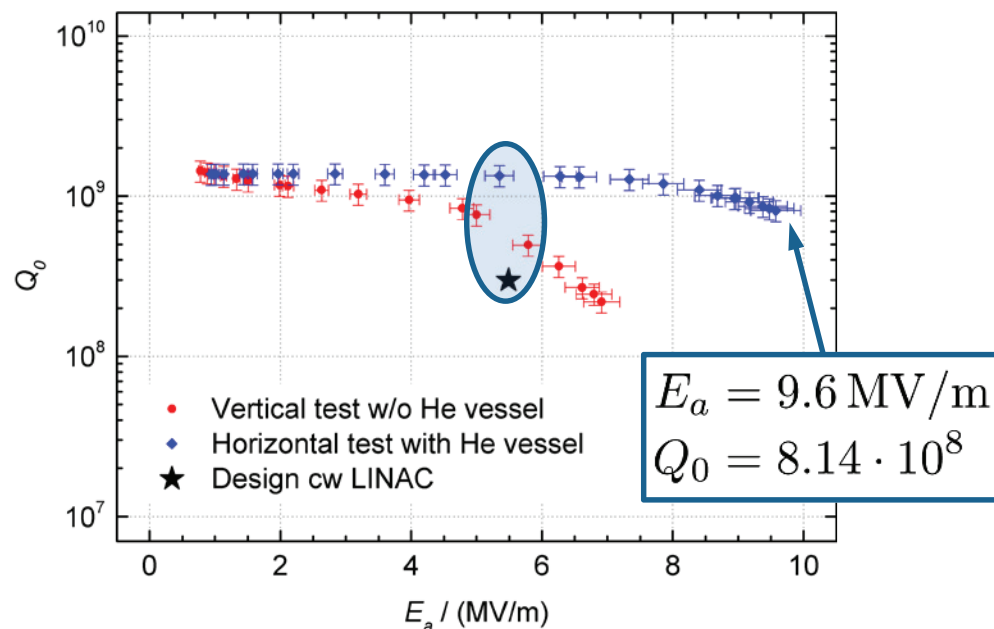
Thermal quenching above 9.6 MV/m

RF test in horizontal orientation with He vessel at 4.2 K



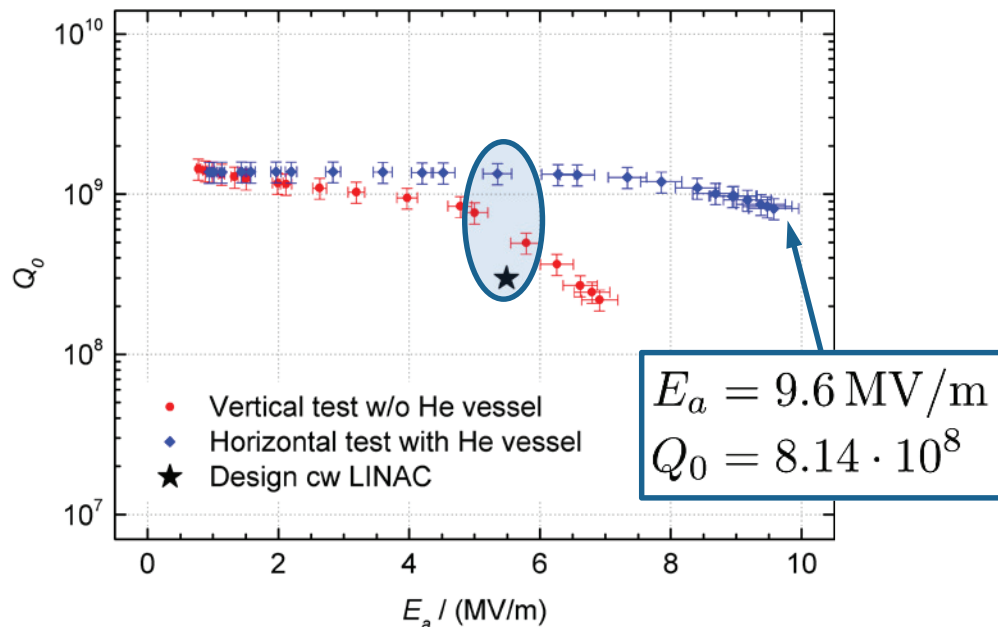
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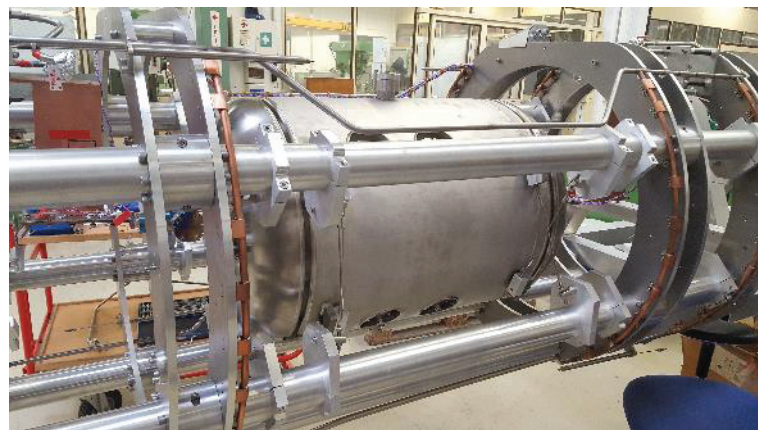
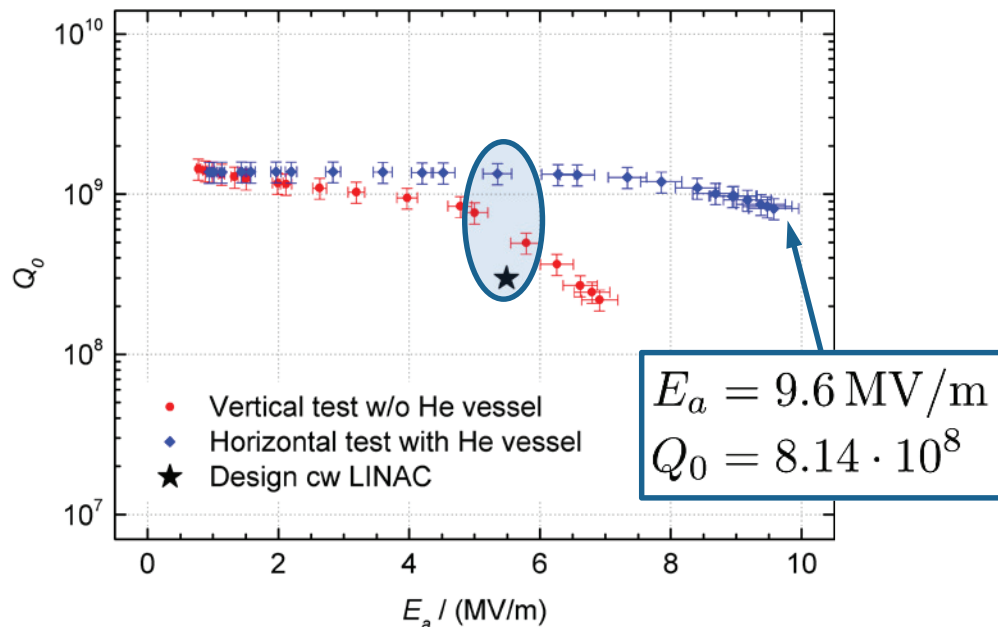
RF test in horizontal orientation with He vessel at 4.2 K



- ➡ Improved performance due to an advanced HPR
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- ➡ Very high gradient
- ➡ Thermal quenching above 9.6 MV/m

		Vertical test w/o He vessel	Horizontal test with He vessel
Q_0^{low}		$1.44 \cdot 10^9$	$1.37 \cdot 10^9$
R_S	nΩ	36	38
R_{BCS}	nΩ	15	15
R_{mag}	nΩ	9	12
R_0	nΩ	12	11
E_a	MV/m	6.9	9.6
Q_0		$2.19 \cdot 10^8$	$8.14 \cdot 10^8$
V_a	MV	4.2	5.9
E_p	MV/m	43	60
B_p	mT	39	55

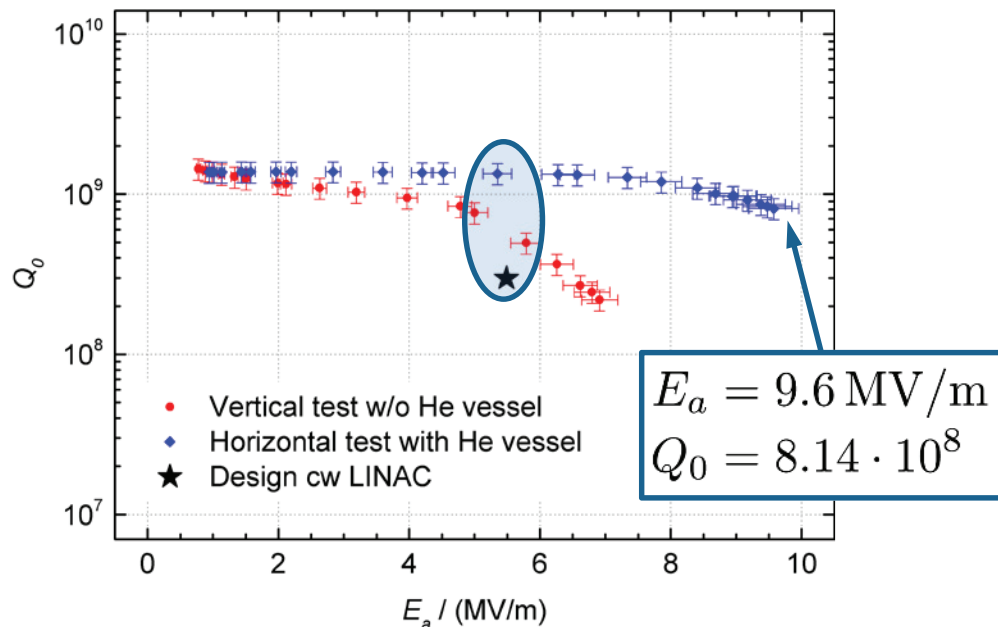
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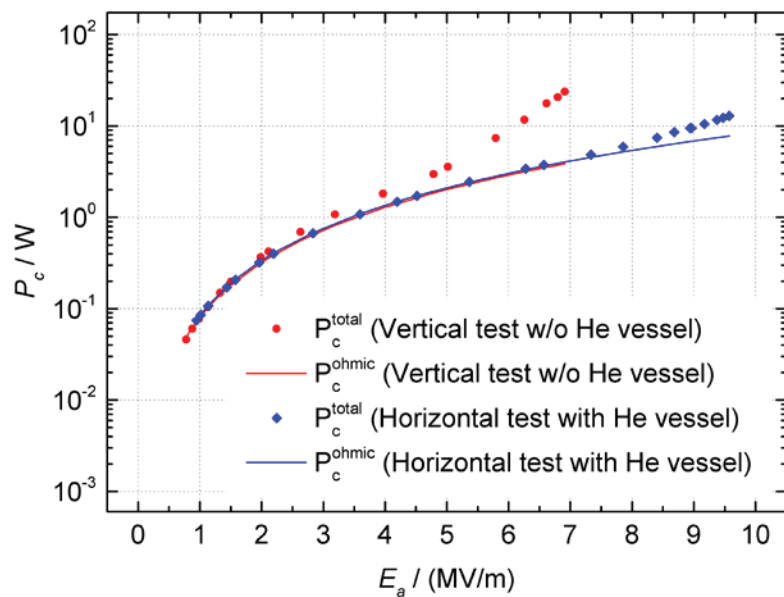
RF test in horizontal orientation with He vessel at 4.2 K



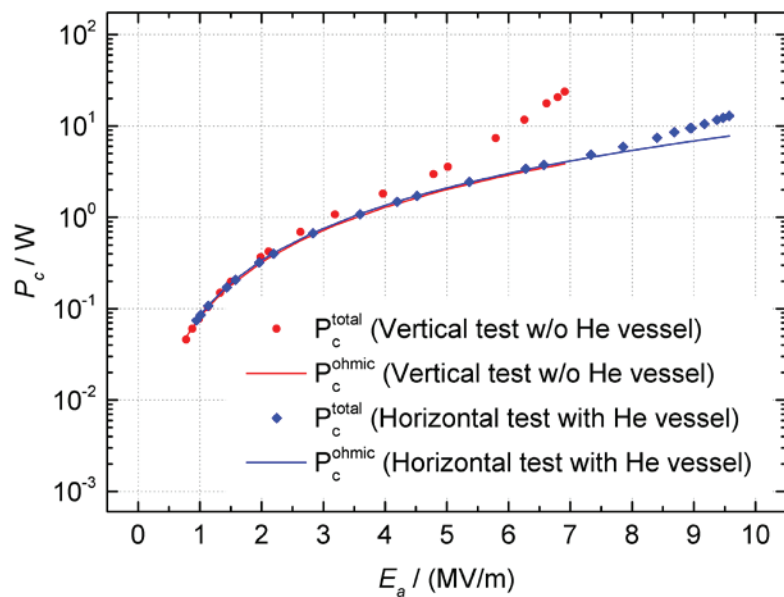
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Total losses inside the cavity

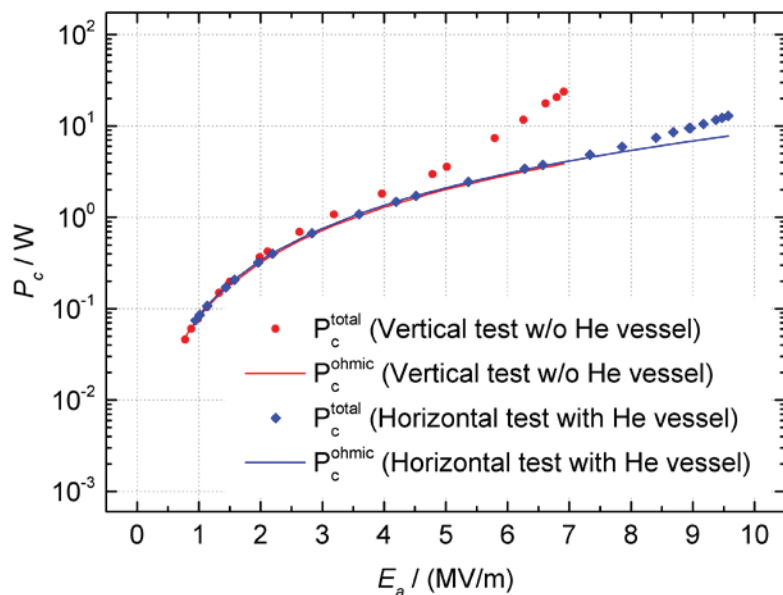


Total losses inside the cavity



First RF test

Total losses inside the cavity



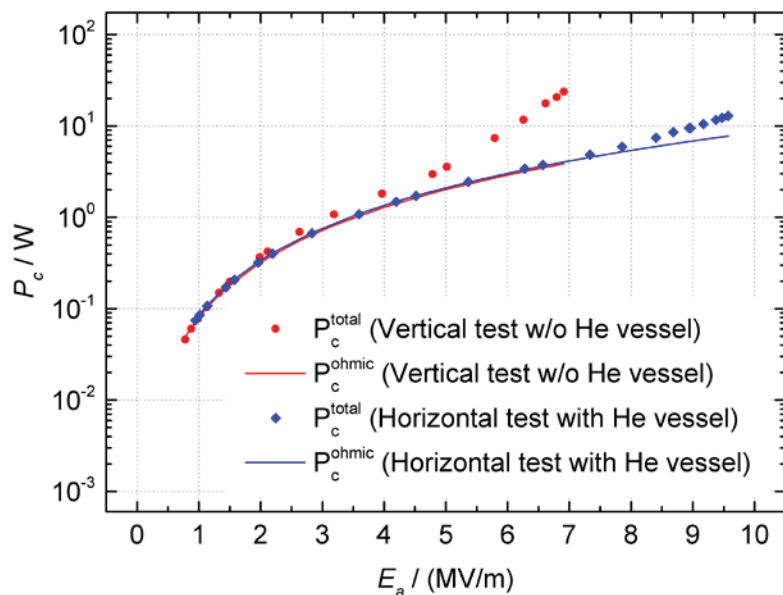
First RF test

Rapidly increase of total losses

at $E_a = 5 \text{ MV/m}$ ($E_p = 32 \text{ MV/m}$)

Activation of field emitters

Total losses inside the cavity



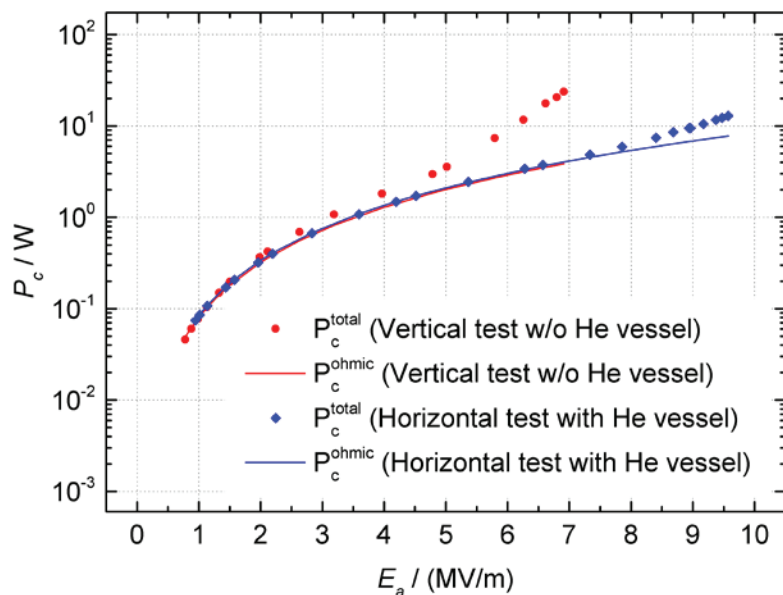
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➡ Rapidly increase of total losses

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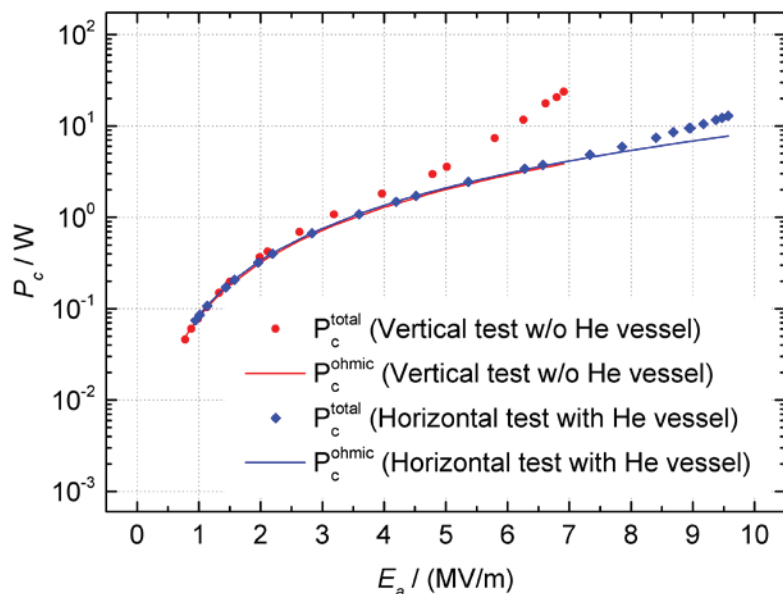
Total losses inside the cavity



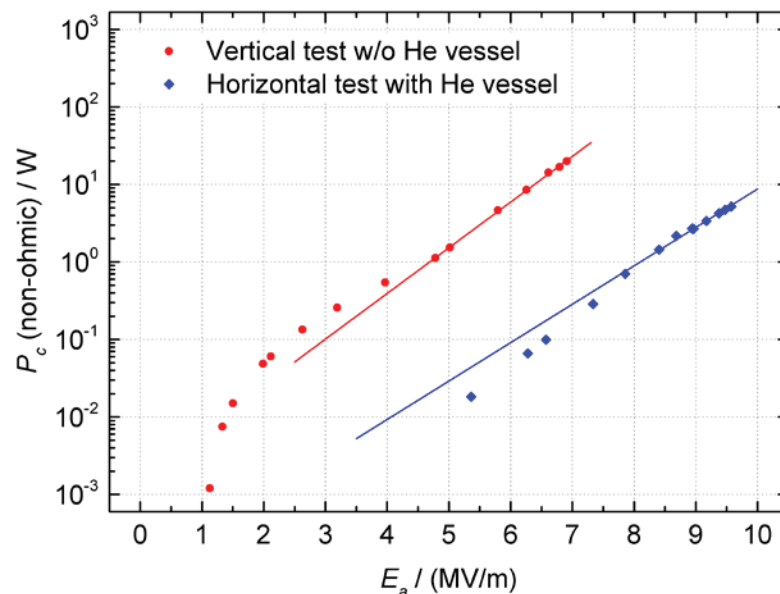
First RF test

- ➡ Rapidly increase of total losses
at $E_a = 5$ MV/m ($E_p = 32$ MV/m)
- ➡ Activation of field emitters

Total losses inside the cavity



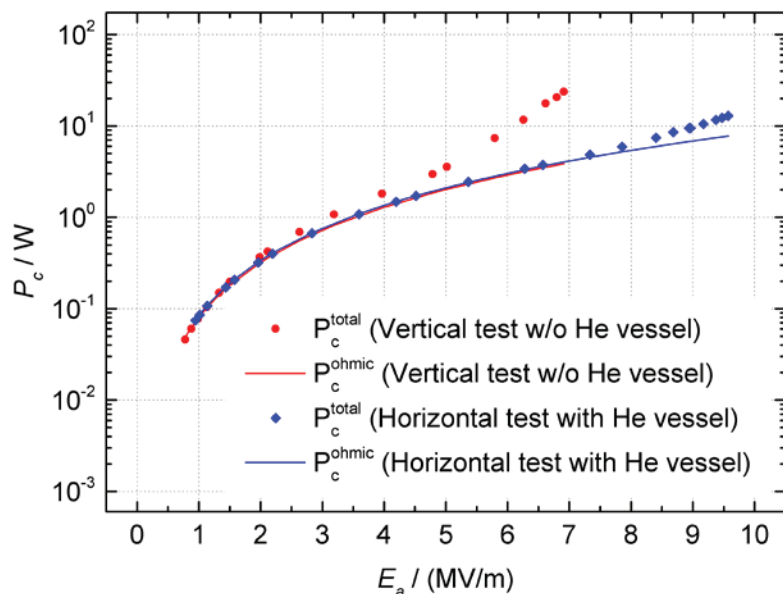
Non-ohmic losses



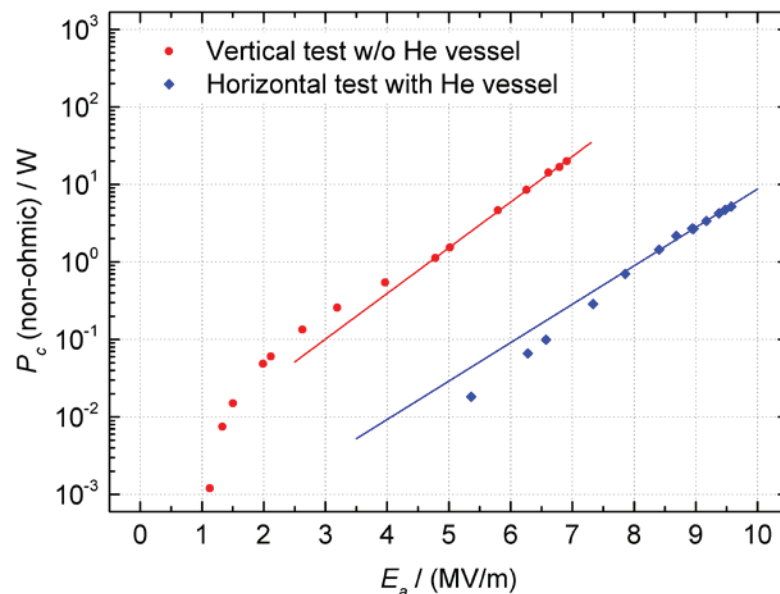
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Total losses inside the cavity



Non-ohmic losses

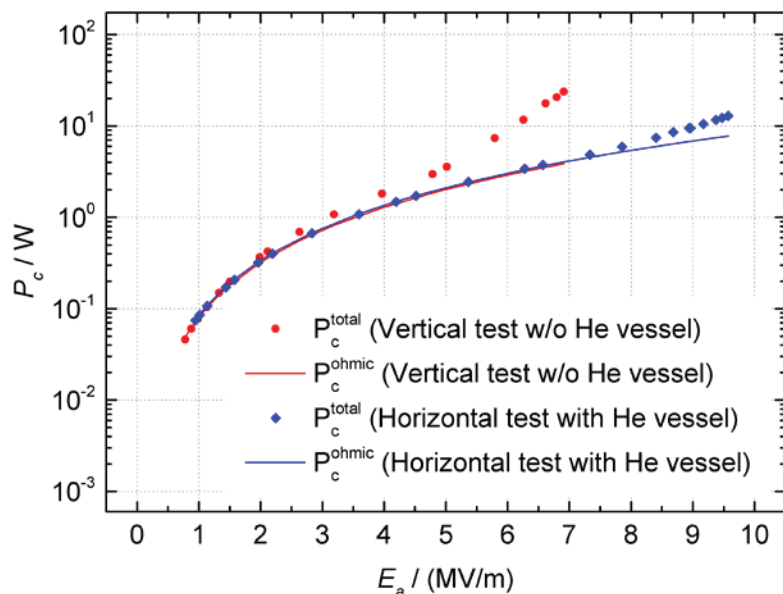


First RF test

- ➡ Rapidly increase of total losses
at $E_a = 5$ MV/m ($E_p = 32$ MV/m)
- ➡ Activation of field emitters

Second RF test (advanced HPR)

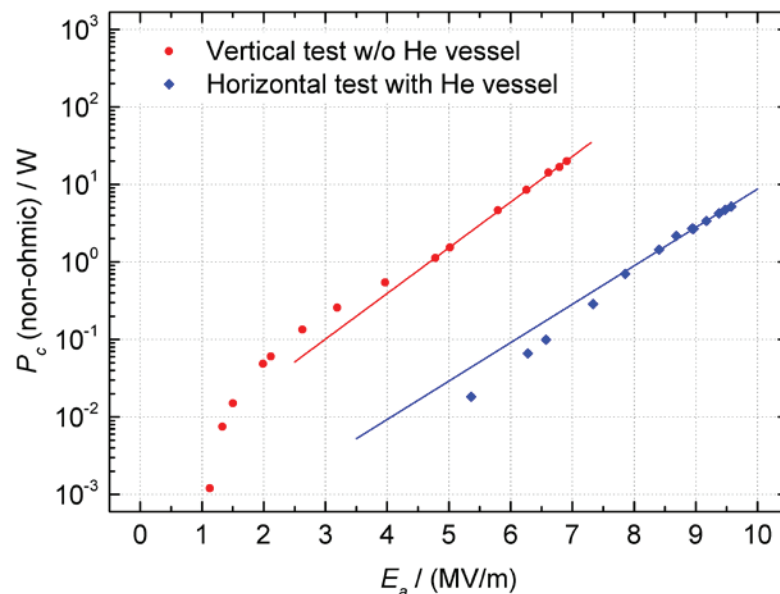
Total losses inside the cavity



First RF test

- ➡ Rapidly increase of total losses
at $E_a = 5$ MV/m ($E_p = 32$ MV/m)
- ➡ Activation of field emitters

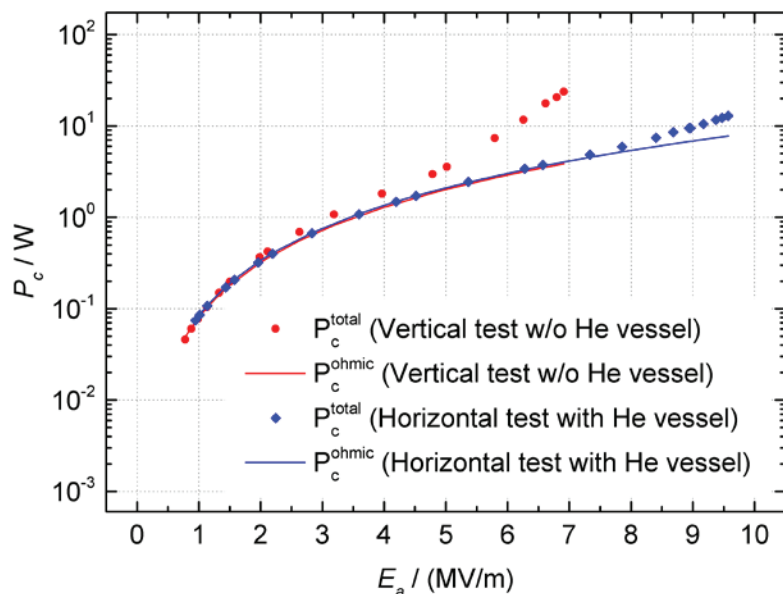
Non-ohmic losses



Second RF test (advanced HPR)

- Extremely reduced total losses
- Minor field emitter activation
at $E_a = 7.8$ MV/m ($E_p = 49$ MV/m)

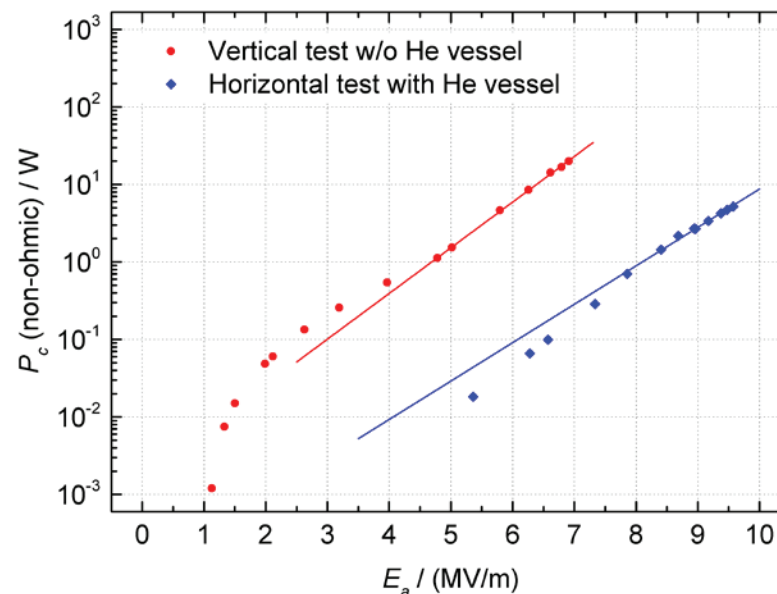
Total losses inside the cavity



First RF test

- ➡ Rapidly increase of total losses
at $E_a = 5$ MV/m ($E_p = 32$ MV/m)
- ➡ Activation of field emitters

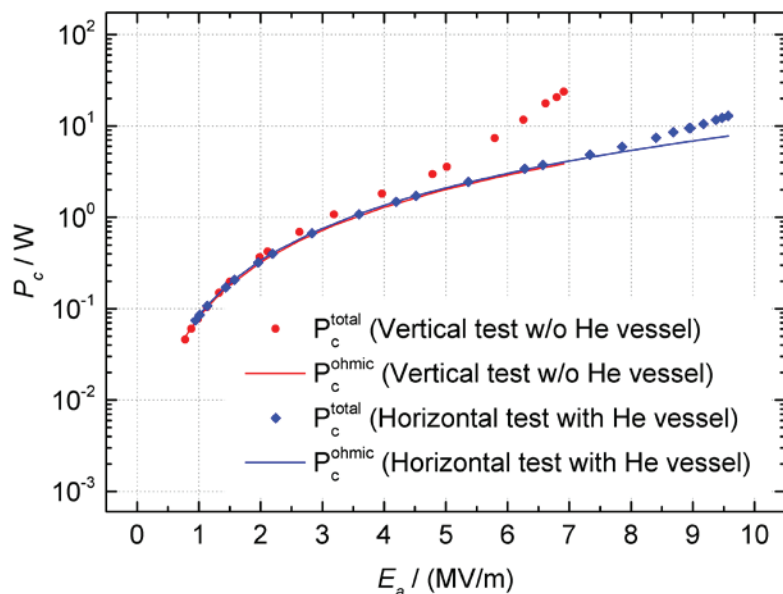
Non-ohmic losses



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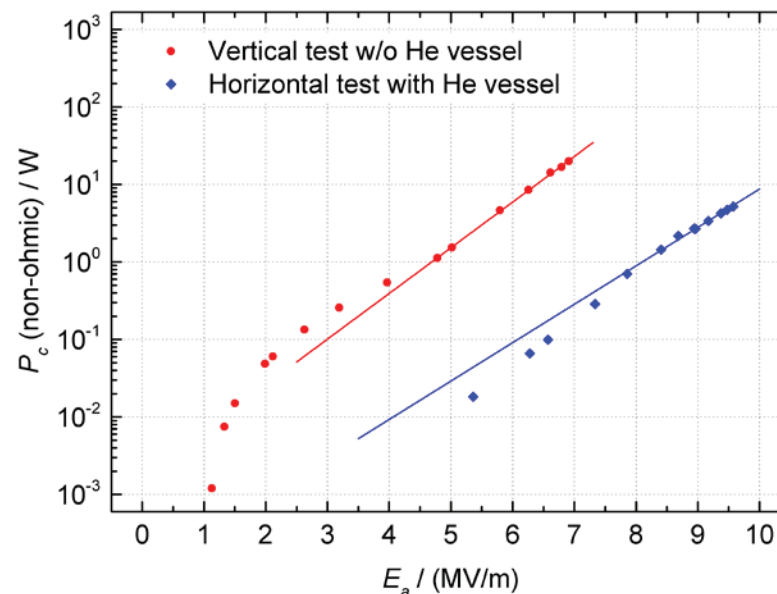
Total losses inside the cavity



First RF test

- ➡ Rapidly increase of total losses
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Non-ohmic losses



Second RF test (advanced HPR)

- ➡ Extremely reduced total losses
- ➡ Minor field emitter activation
at $E_a = 7.8$ MV/m ($E_p = 49$ MV/m)

09/2012

Project start
Ordering of the cavity, solenoids and 5 kW cw RF amplifier

03/2014

Intermediate RF measurements on the cavity during fabrication

12/2015

Commissioning of cryomodule and solenoids

01/2016

First RF test of the cavity at IAP (vertical orientation)

09/2016

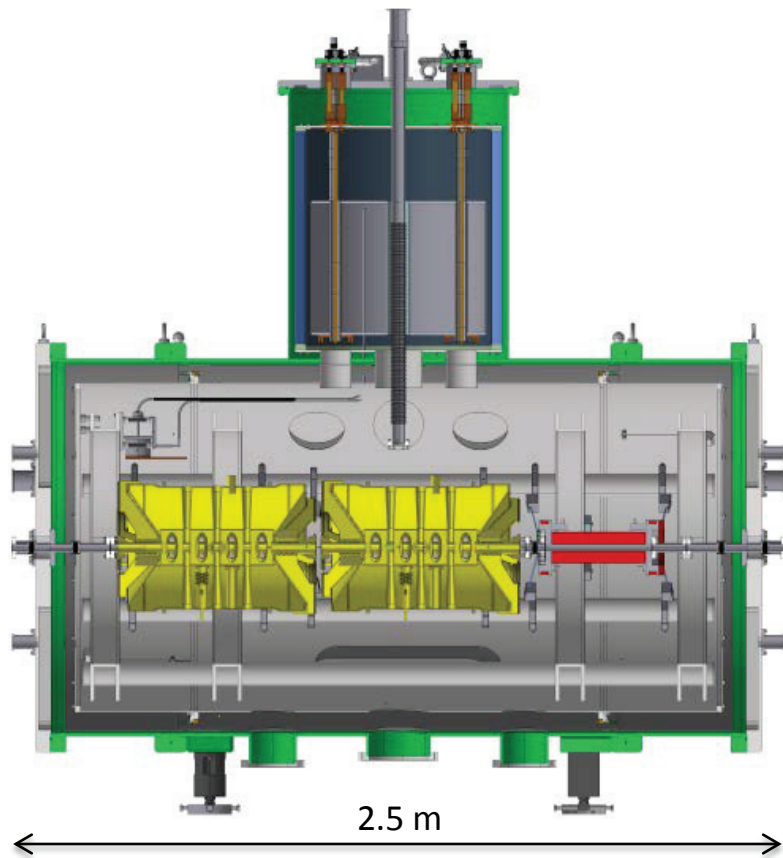
Assembly of He vessel to the cavity and HPR

10/2016

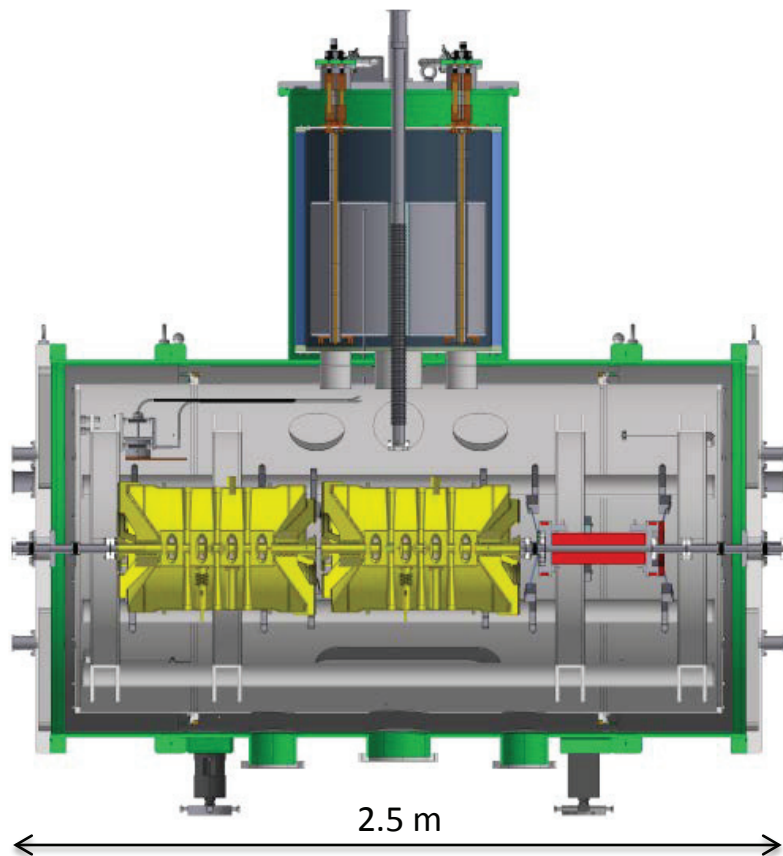
Second RF test of the cavity at GSI (horizontal orientation)

03/2017

Beam test of the demonstrator



Further approach



- New standard cryomodule layout containing 2 short CH cavities and 1 solenoid
- 3 new cryomodules + existing demonstrator build the “Advanced demonstrator”
- Advanced demonstrator allows first experiment at coulomb barrier for light and medium heavy ions
- Simplified cavity design
(easier manufacturing & surface processing)
- Cavity 2 & 3 are already in production
(delivery at 4th quarter of 2017)
- Tendering for cryostat at 2nd quarter of 2017

- Cryomodule and solenoids ($B = 9.3$ T) successfully tested at 4.2 K
- Demonstrator cavity was vertically tested w/o He vessel at IAP
- He jacket was assembled to the cavity and HPR was improved
- Successful horizontal RF test of cavity with He vessel at GSI
- Very promising gradient of 9.6 MV/m could be reached
- Production of further two short CH cavities already started
- A new standard cryomodule layout was defined

- RF conditioning of 5 kW cw power couplers
- Commissioning of local clean room at GSI
- Beam test of the demonstrator (CH cavity & solenoids)
- Further improvement of the demonstrator cavity
 - Rinsing each quadrant of the cavity with optimized HPR device
 - Argon plasma discharge
 - 300°C bakeout
- Delivery of short superconducting CH cavities
- Commissioning of infrastructure at HIM
 - Cleanroom environment
 - High Pressure Rinsing
 - Buffered Chemical Polishing (optional)
 - Cavity bakeout
 - Setup for cavity RF testing

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 - Cavity bakeout
 - Setup for cavity RF testing



Thank you very much for your attention!

Collaboration partners

GSI

A. Rubin
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IAP/GU Frankfurt

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